Introduction to Biomaterials

Lecture [9]

Clinical applications of Biomaterials

Goals

- Advantages of biomaterials in clinical applications
- Why different biomaterials are selected for different medical uses
- The role of biomedical engineers in medicine

Biomaterials in Modern Healthcare

- Biomaterials are increasingly used inside the human body. They may be:
  - Synthetic materials
  - Natural materials
  - Temporary devices
  - Permanent implants
- Examples include:
  - Pacemakers
  - Heart valves
  - Joint replacements
  - Dental implants



Requirements Before Clinical Use

- Any biomaterial used in patients must meet **strict requirements.**
- Important criteria include:
  - Suitable material properties
  - Safe preparation and processing
  - Proper sterilization
  - Biocompatibility
  - Short-term safety
  - Longterm performance
- **Being useful** in **engineering does not** automatically mean the **material** is **safe** for the **body**.

Why Biomaterials Are Used Clinically

Treat

- Help control or manage a disease

Repair

- Restore the function of damaged tissue or organs

Restore

- Rebuild or maintain body shape and appearance

Examples of Clinical Use

Type	Meaning	Examples
Treatment	Helps treat a specific disorder	Pacemakers, cardiac valves
Repair	Restores damaged function	Joint replacements, fracture fixation devices
Restoration	Restores shape or esthetics	Dental implants, breast implants

Beyond Survival: Patient Needs

- Clinical biomaterials are not used only to save life.
- They may also improve:
  - Rehabilitation
  - Comfort
  - Convenience
  - Appearance
  - Psychological wellbeing
  - Social confidence
- **Example:** A dental implant restores chewing function, but also improves appearance and confidence.

1. Cardiovascular Assist Devices

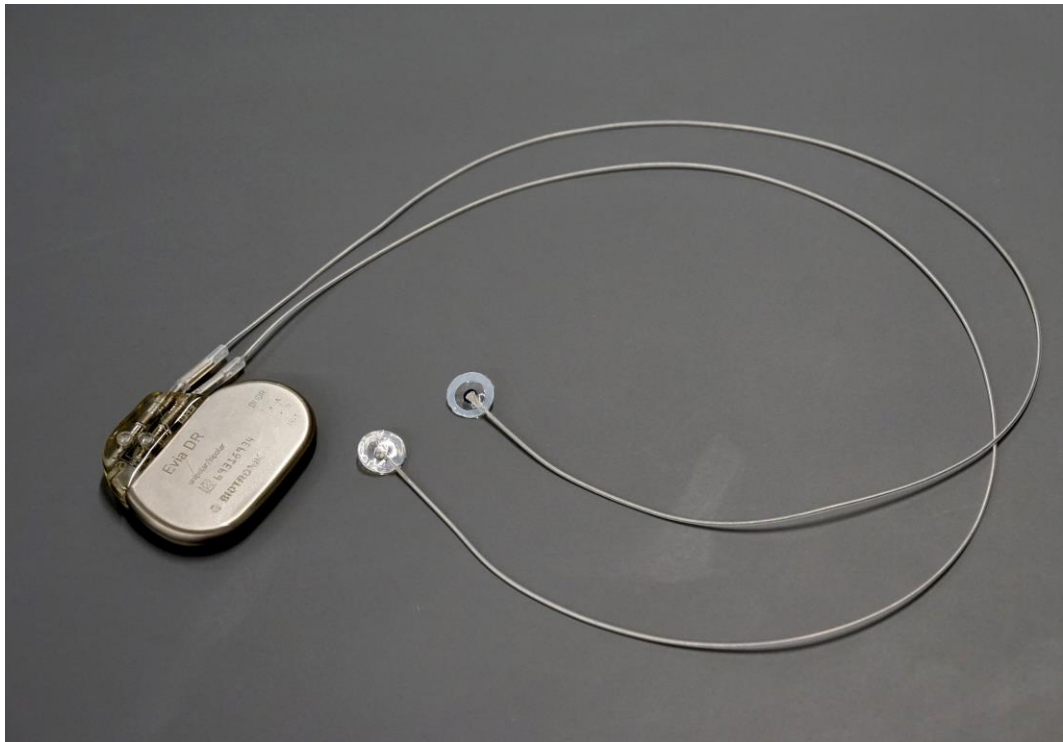
1. Cardiovascular Assist Devices

- **Cardiac assist** devices include devices such as **pacemakers**.
- In these devices, **cardiac pacing leads** must **survive** inside the **harsh endocardial environment**.
- The leads are positioned in the heart and connected to a battery powered heart failure device.

Function of Pacing Leads

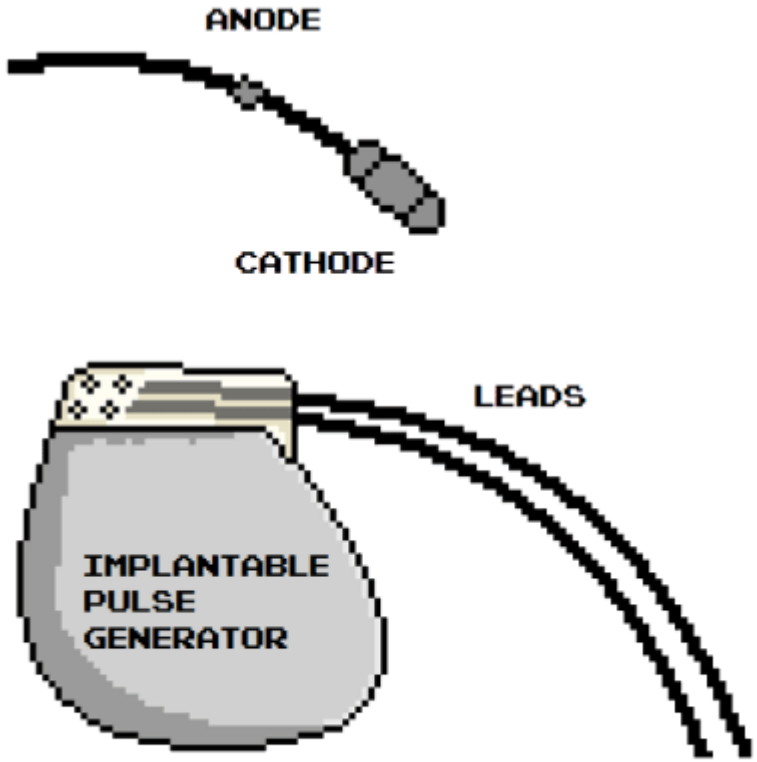
- **Pacing leads** allow **transmission** of electrical **signals** to the **heart**.
- This is important for cardiac **resynchronization therapy**.

- The lead must help the device:
  1. Deliver electrical signals to the heart
  2. Sense cardiac signals from the heart



Main Components of Pacing Leads

- **Major components** of cardiac pacing leads include:
  1. Electrode
  2. Conductor
  3. Insulation
- Each component is critical to pacemaker function.
- Therefore, material selection for each component is important.



Pacing Lead Failure

- **Most pacing failures** occur because of **malfunction** of the **pacing leads**.
- This is because the **leads** are **exposed** to the harsh **endocardial** environment.
- Some metals such as **copper**, **zinc**, **nickel**, and **silver** can cause **toxic reactions** with the **myocardium**.

Electrode Materials

- **Current electrodes** are made from materials such as:
  - Platinum–iridium
  - Elgiloy
  - Carbon



- **Elgiloy** is an alloy containing **nickel**, **chromium**, **cobalt**, **manganese**, **molybdenum**, and **iron**.

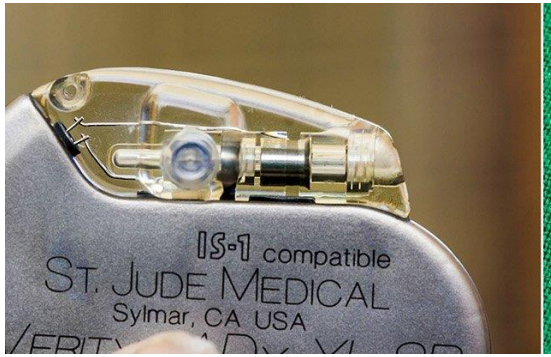
Ideal Electrode Properties

- The ideal electrode should:
  - Be biologically inert
  - Resist degradation
  - Resist corrosion
  - Cause minimal tissue inflammation
  - Cause minimal fibrosis at the electrode-endocardial interface

- **Steroid releasing electrode** designs can reduce inflammatory response after implantation.

Pacing Lead Conductor

- The **pacing lead conductor** is a **metal wire**.
- **Its function is to:**





  - Conduct current from the pulse generator to the pacing electrode

    Send cardiac signals from the electrode to the sensing amplifier

- The **conductor** must support both **stimulation** and **sensing**.

Ideal Conductor Properties

- The **pacing lead conductor** should ideally be:
  - Noncorrosive
  - Strong enough in tension
  - Resistant to metal fatigue

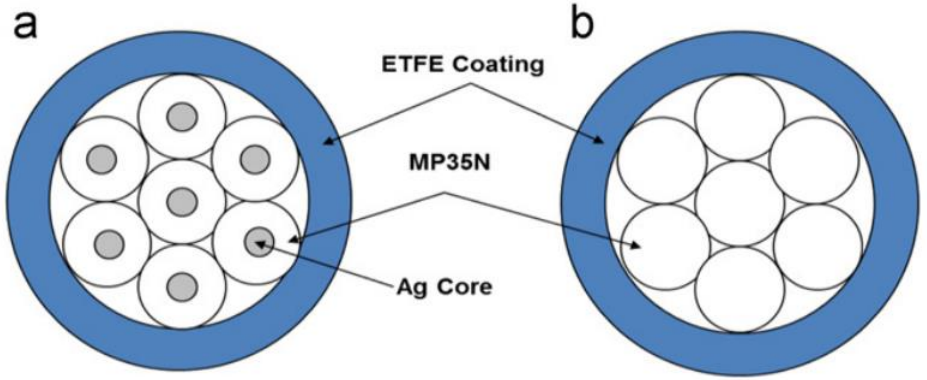


Fig. 3 – Structure of 1 × 7 Ag/MP35N and 1 × 7 MP35N cables. (a) 1x7 Ag/MP35N cable. (b) 1x7 MP35N cable.

- Current conductor materials include noncorrosive alloys such as **MP35N**.
- **MP35N** contains nickel, chromium, cobalt, and molybdenum.

Insulating Materials

- **Insulating materials** are used to **separate** the **conductor coils**.

- **Common biomaterials** used as insulators in cardiac pacing leads include:
  - Silicone rubber
  - Polyurethane

- These materials protect the conductor and help maintain lead function.

Silicone Rubber as an Insulator

- **Silicone rubber** is partly organic and partly inorganic.

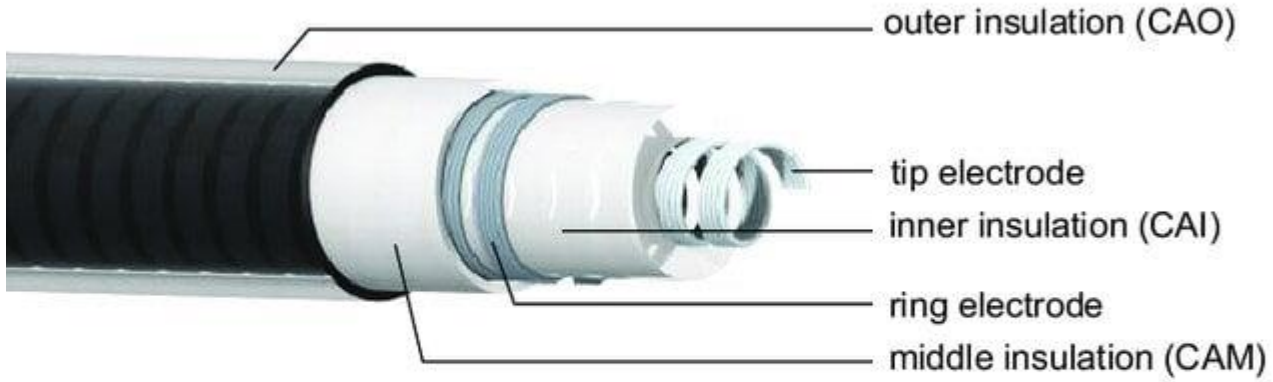
- Its main polymer chain contains alternating silicon and oxygen atoms.
- Important properties include:
  - Biocompatibility
  - High lubricity
  - Excellent resistance to water
  - Chemical inertness
  - Ability to retain physical properties at elevated temperatures

Polyurethane as an Insulator

- Important properties of polyurethane include:
  - Loadbearing capacity comparable to cast steel
  - Exceptionally smooth surfaces
  - Resistance to fungal colonization
  - Excellent hydrolytic stability
- However, polyurethane insulators may fail because of crack propagation and metal ion oxidation.

Insulating Materials in Cardiac Pacing Leads

- Used to **separate and protect conductor coils**.
- Main materials:
  - **Silicone rubber**
  - **Polyurethane**
- **Silicone rubber:**
  - Biocompatible
  - Water resistant
  - Chemically stable
- **Polyurethane:**
  - Stronger and smoother
  - Good stability
  - May fail by cracking or oxidation



Coaxial lead design		Insulation material
Outer insulation layer	(CAO)	Pellethane 2363 55D
Middle insulation layer	(CAM)	Silicone MED 4765
Inner insulation layer	(CAI)	Silicone MED 4765

Silicone Rubber vs Polyurethane

Biomaterial	Advantages	Disadvantages
Silicone rubber	Easily fabricated; easily moldable; easily repairable; long history of use	Tears easily; cuts easily; thrombogenic; higher friction coefficient in blood; subject to cold failure
Polyurethane	High tear strength; high tensile strength; resistant to cuts; low friction coefficient in blood; high abrasion resistance; high compressive properties; less thrombogenic	Relatively stiff; lacks repairability; sensitive to manufacturing process; possible environmental stress cracking; possible metal ion oxidation



Figure 14.1

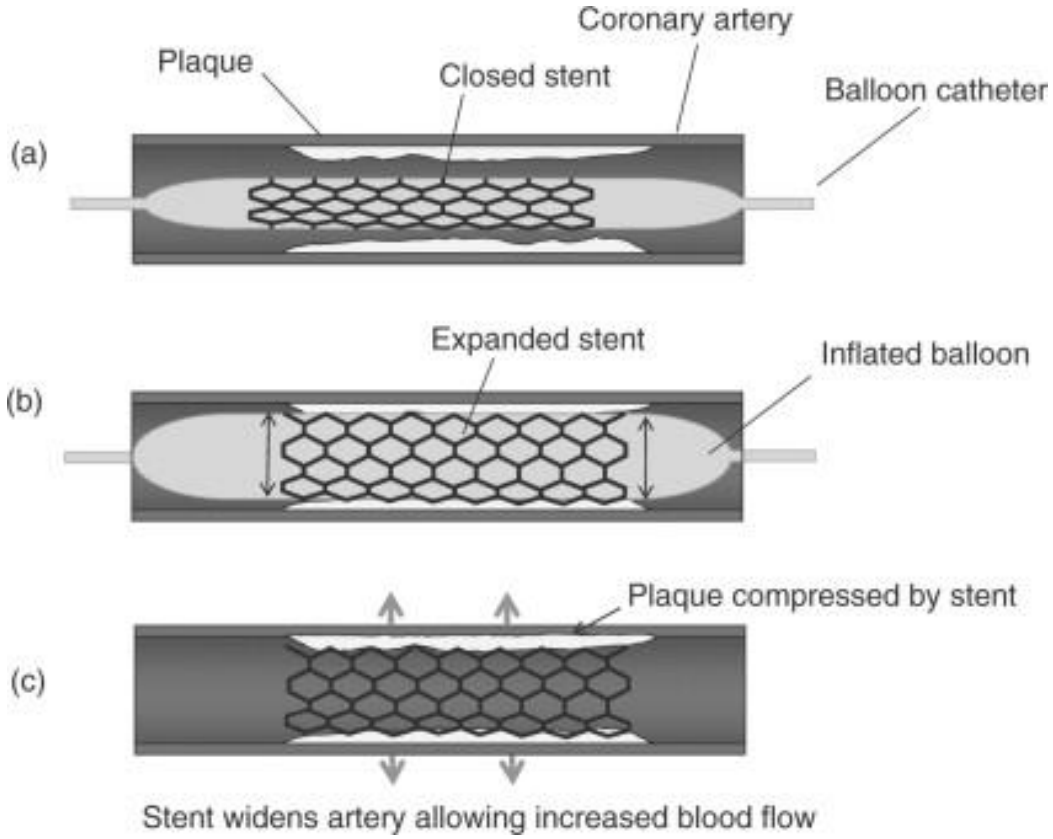
Photograph of EASYTRAK®, EASYTRAK®2, and EASYTRAK®3 left-ventricular coronary venous pacing/sensing leads. (Courtesy of Boston Scientific, Natick, MA, USA.)

2. Cardiovascular stents

2. Cardiovascular Stents

- Cardiovascular stents are tubular scaffolds.

- Bare metal stents are commonly made from:
  - 316L stainless steel
  - Nitinol / NiTi alloy
  - Cobalt–chromium-based alloys



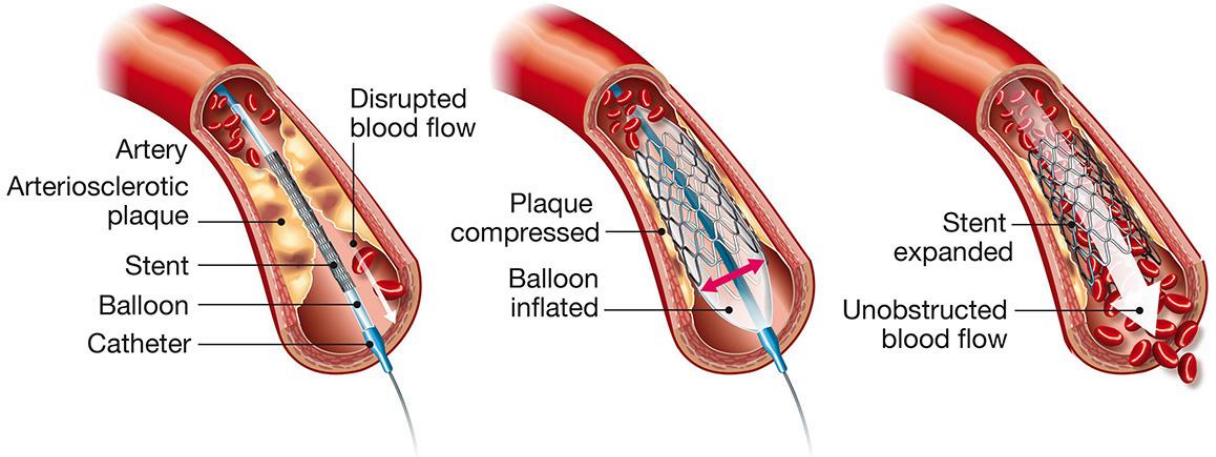
Why Stent Material Matters

- **Different metals** have **different properties**.
- These differences strongly influence **stent design** and **performance**.

- Examples:
  - Nitinol can be used for selfe-xpandable stents
  - Cobalt–chromium stents can have thinner struts than stainless steel
  - Cobalt–chromium stents are less prone to biocorrosion than stainless steel

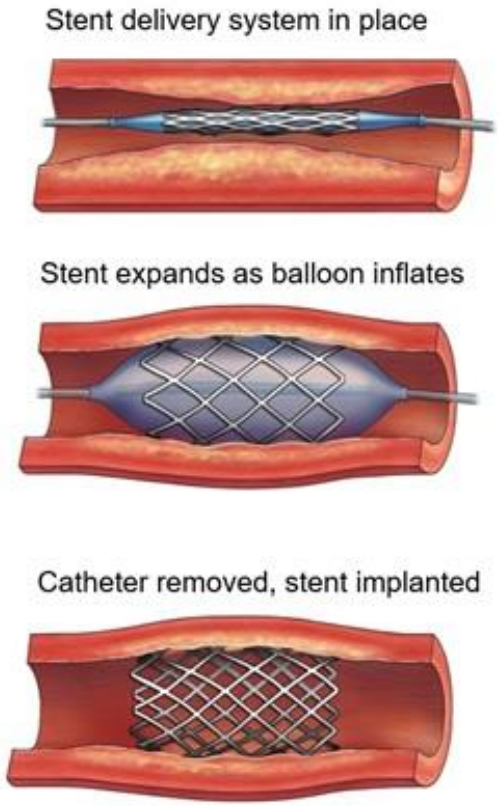
Clinical Use of Stents

- Stents are used to reduce coronary artery blockage.
- They are used during revascularization procedures of:
  - Coronary arteries
  - Peripheral arteries
- Their main purpose is to keep the vessel open and improve blood flow.



Coronary Stenting Procedure

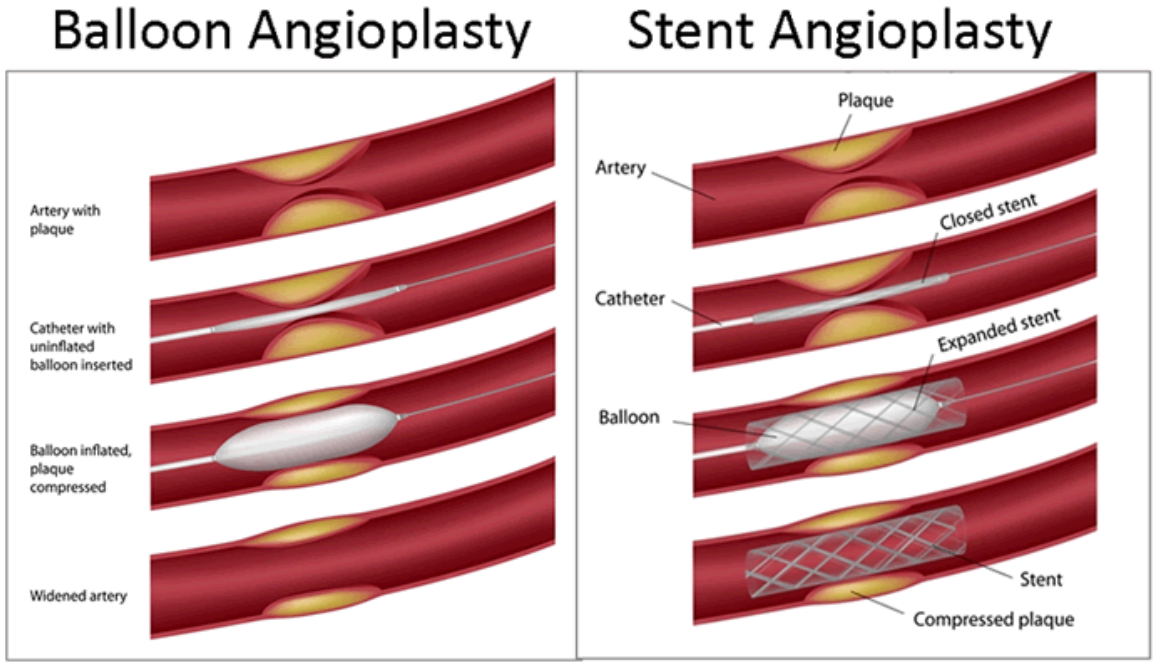
- A stent is mounted on a balloon catheter in a crimped or collapsed state.
- During the procedure:
  - The catheter reaches the blocked artery
  - The balloon is inflated
  - The stent expands
  - The stent pushes against the artery lumen
  - The blocked area becomes dilated
- After this, the balloon is deflated and removed.



Stenting vs Balloon Angioplasty

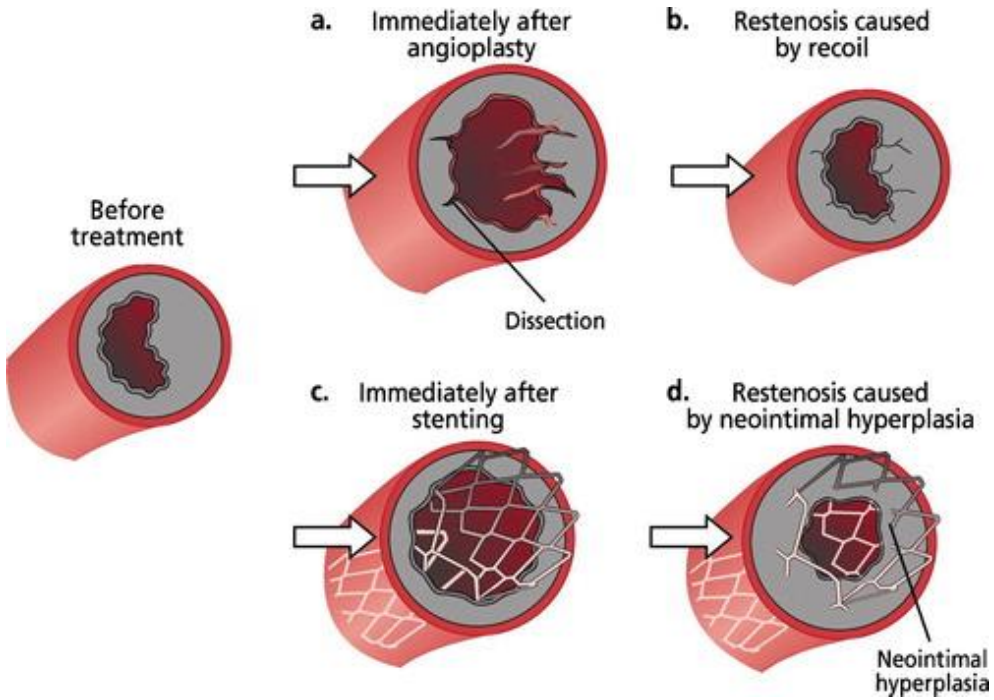
- In conventional balloon angioplasty:
  - The balloon dilates the blocked artery
  - The balloon is removed after dilation

- In coronary stenting:
  - The balloon expands the stent
  - The stent remains inside the artery
  - The stent helps keep the vessel open



Restenosis

- **Restenosis** means **renarrowing** of the blood vessel.
- It is defined as more than *50% renarrowing* of vessel diameter **at 6 months' followup.**
- It can occur because of:
  - Smooth muscle cell hyperproliferation
  - Thickening
  - Scarring
  - Remodeling of the vessel region



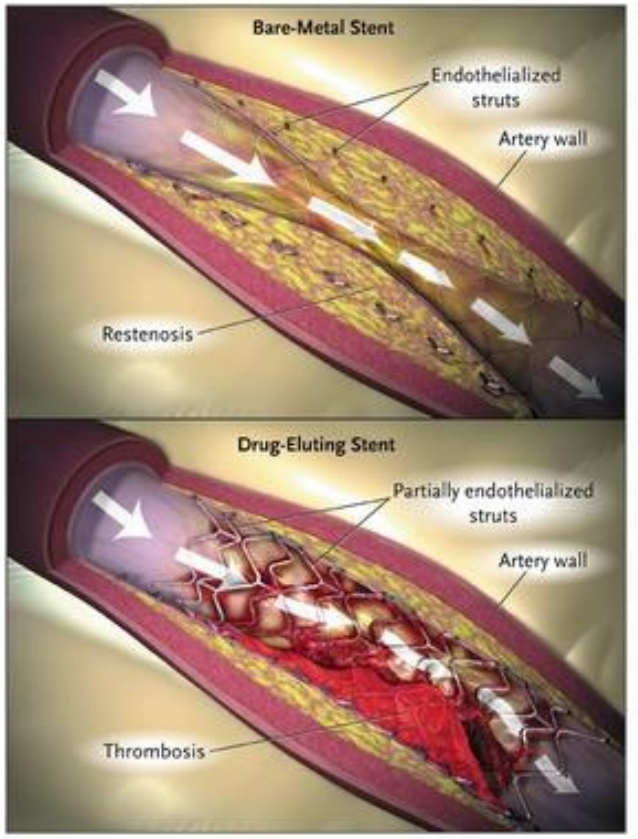
Benefits of Coronary Stenting

- Compared with conventional balloon angioplasty, stenting can reduce:
  - Restenosis
  - Elastic recoil of the coronary artery
  - Neointimal hyperplasia

- Because of these benefits, stenting is now a wellestablished technique in endovascular interventions.

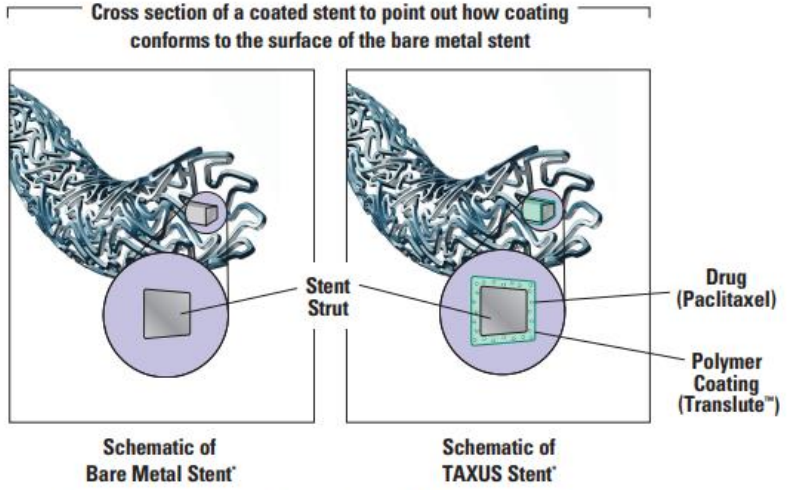
Limitation of Bare Metal Stents

- Bare metal stents reduce the problem of elastic recoil.
- However, some problems may still occur in some patients:
  - Instent restenosis
  - Neointimal hyperplasia
- This led to the development of **drug eluting stents**.



Drug Eluting Stents

- **Drug eluting stents** are designed to slowly release a drug at the implantation site.
- The drug is released over time to **pharmacologically reduce instent restenosis**.
- Examples mentioned:
  - TAXUS® paclitaxeleluting coronary stent
  - CYPHER® sirolimuseluting stent



Note: A green color is used to show coating but actual coating is clear.
**Liberté® stent design shown*

TAXUS Paclitaxel-Eluting Coronary Stent Systems:
TAXUS® Express²™ Paclitaxel-Eluting Coronary Stent System
TAXUS® Liberté® Paclitaxel-Eluting Coronary Stent System

Polymeric Coatings in Drug Eluting Stents

- **Biodegradable polymers** may be used as **matrices** for **drugeluting coatings**.
- A common example is:
  - PLGA: poly (lacticcoglycolic acid)
- PLGA is a block copolymer of:
  - PLA
  - PGA
- **The polymer coating helps carry and release the drug.**

Problems with Drug Eluting Stents

- Current drug eluting stents still have some concerns.
- Possible problems include:
  - **Nonuniform drug release** from **polymeric** coatings
  - **Allergic reactions** to stent **coatings** in some patients

- Because of this, new coating biomaterials are being investigated.

Other Stent Applications

- Stents are **not limited** to **blood vessels**.

- They can also be used to **enlarge hollow structures** such as:
  - Tracheobronchial tree
  - Biliary duct
  - Urinary tract

- So, the general idea of a stent is to support and keep a lumen open.

(a)

(b)

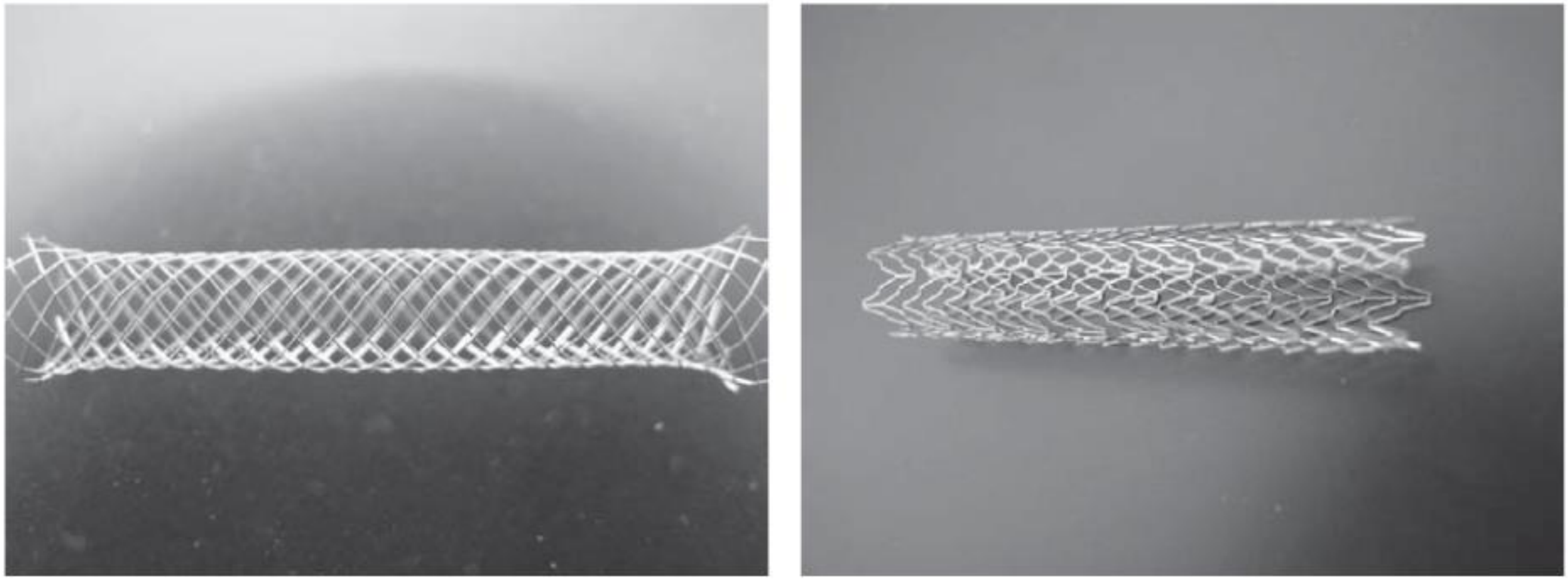


Figure 14.2

Photographs of metal stents with different strut designs.

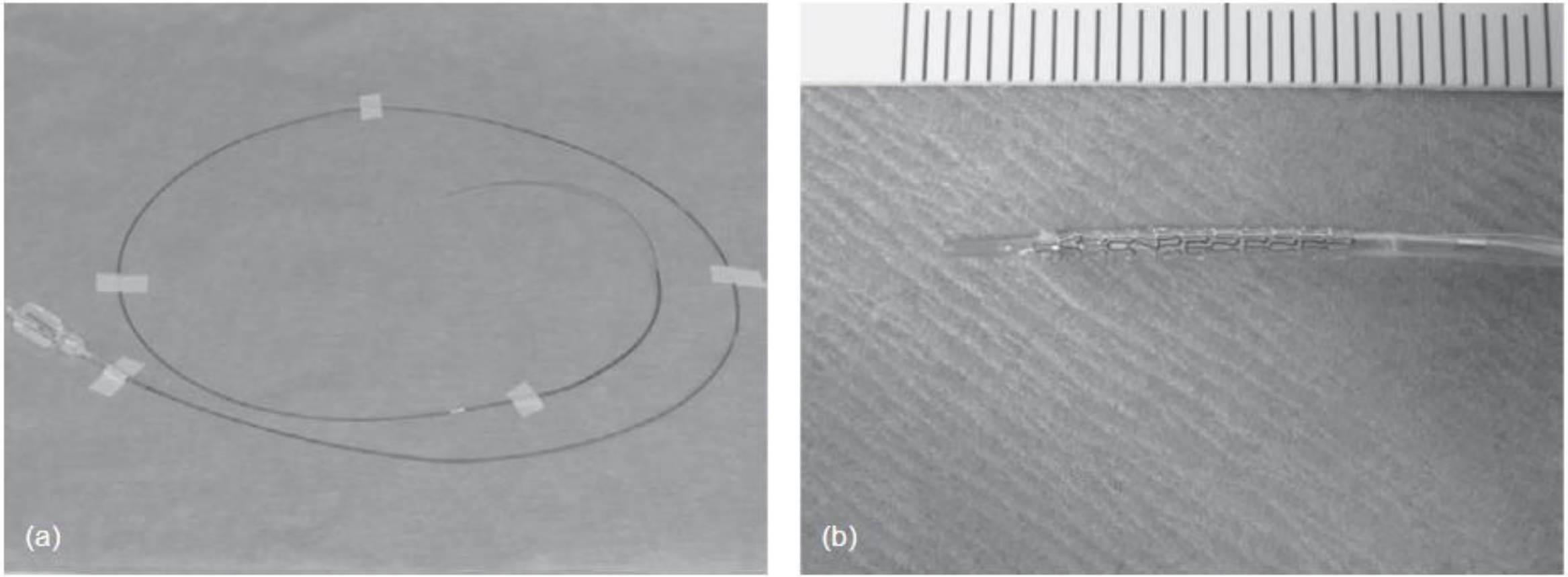


Figure 14.3

Stent on a catheter in a "crimped" or collapsed state. (a) Entire catheter. (b) Enlarged view at the end of the catheter showing the "crimped" stent.

3. Dental restoration

3. Dental Restoration



- Dental restorative biomaterials are used to **restore damaged teeth**.
- Main applications include:
  - **Block replacement** (fillings)
  - **Shell coverage** (crowns)
  - **Surface lamination**
- Main material groups:
  - **Metal alloys**
  - **Ceramics**
  - **Polymers**





Metal Restorative Materials

- Common metal restorative materials include:
  - **Silver amalgam**
  - **Gold**
  - **Stainless steel**
  - **Cobalt–chrome**
- **Silver amalgam:**
  - Commonly used for **posterior teeth**
  - Used where **esthetics are less important**
  - Made mainly from **silver, tin, and copper**
  - Formed by mixing the alloy with **mercury**



Esthetic Restorative Materials

- Esthetic materials were developed to improve **tooth appearance**.
- Main examples:
  - **Resin composite**
  - **Glass ionomer cement**
- These materials are more **tooth-colored** and are commonly used for **anterior teeth**.
  - But is **brittle** and **weaker** than many other restoratives





Esthetic Restorative Materials

- Key limitations:
  - Resin composite:
    - Shrinks during polymerization
    - May form marginal gaps
    - May lead to **secondary caries**
    - Wears more in high load-bearing areas
  - Glass ionomer cement:
    - Adhesive restorative material
    - Has favorable thermal expansion





Other Restorative Materials / Crowns



- Other restorative materials include:
  - **Porcelain**
  - **Cast metal alloys**
  - **Ceramics**
- These are often used for **shell coverage restorations (crowns)**.
- Examples:
  - **Porcelain crowns**
  - **Gold crowns**
- Important note:
  - **Porcelain** is effective, but it is **harder than tooth structure**
  - It may **wear the opposing tooth**
- These restorations are usually **more expensive** because they require **more treatment time**


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Figure 14.4

A dental patient with cavities in the oral cavity filled with amalgam. (Courtesy of Dr. Daniel C. N. Chan, University of Washington, School of Dentistry.)

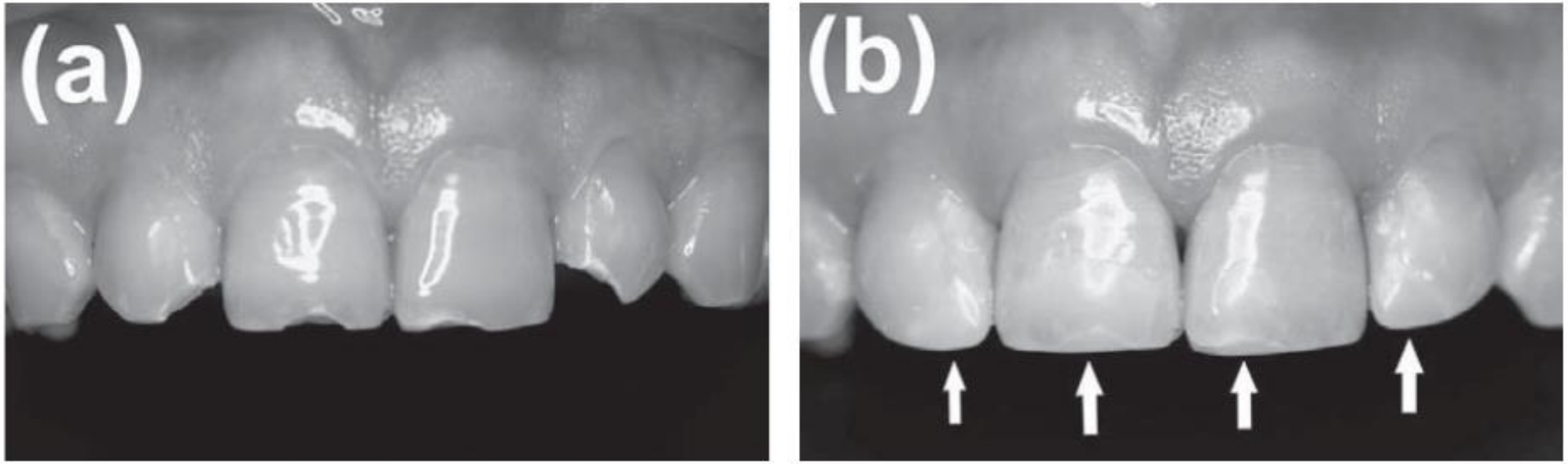
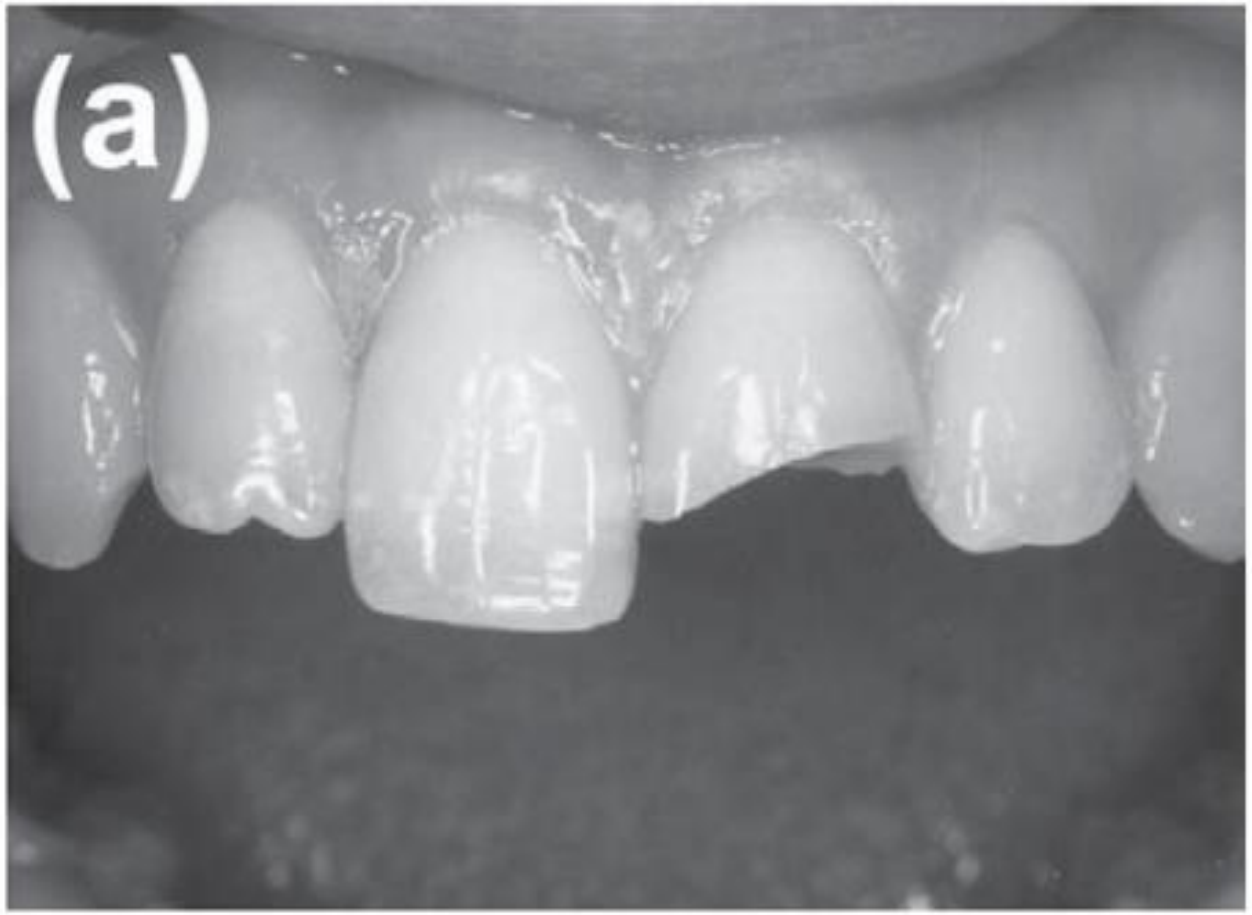


Figure 14.5

A dental patient with (a) chipped incisors, and (b) the incissors filled with resin composite, as indicated by arrows. (Courtesy of Dr. Daniel C. N. Chan, University of Washington, School of Dentistry.)



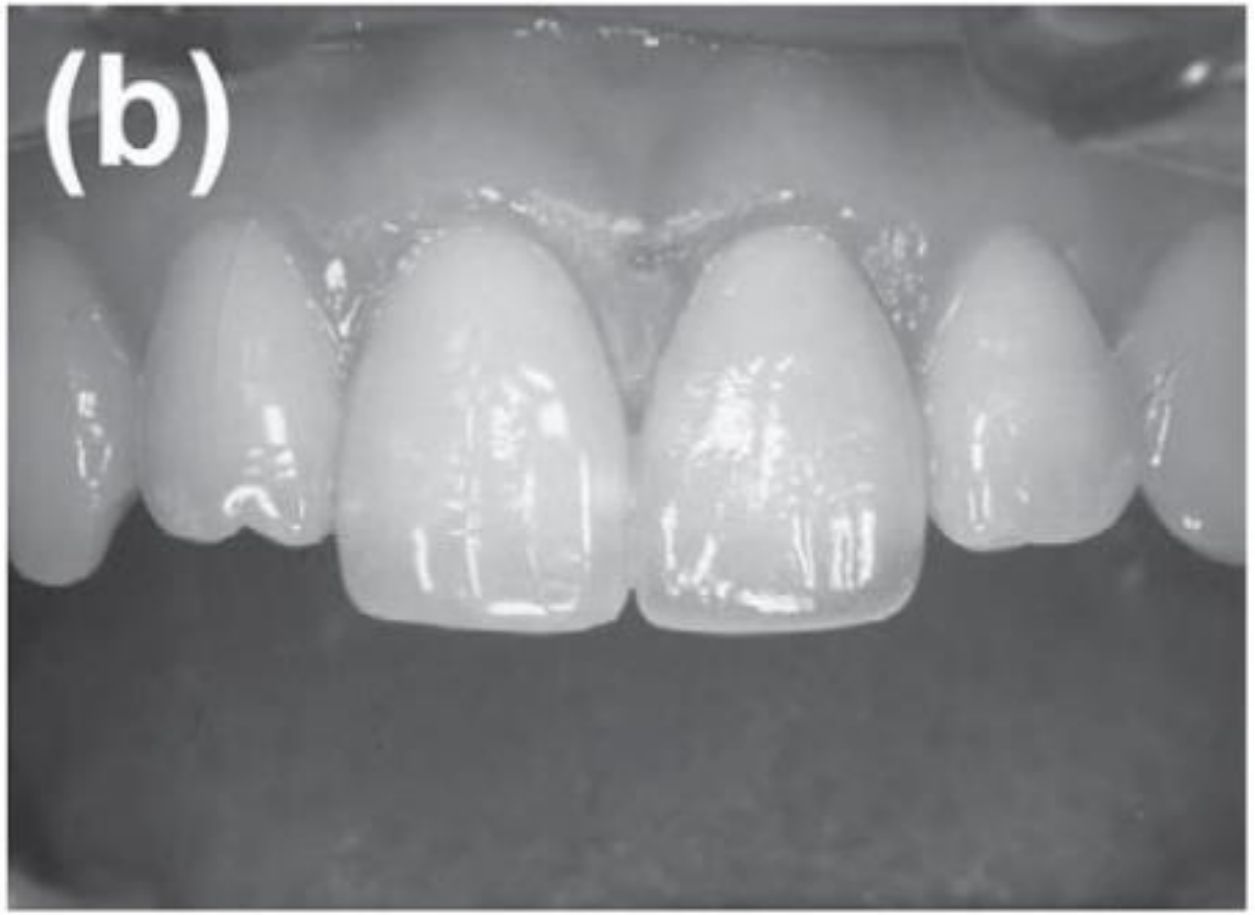


Figure 14.6

A dental patient with (a) a fractured incisor in the maxilla, and (b) the fractured incisor capped with a porcelain crown. (Courtesy of Dr. Namsik Oh, Inha University Hospital, South Korea.)

4. Dental implants

Dental Implants

- Dental implants are used to restore tooth function.
- They are used for:
  - Partially edentulous patients
  - Totally edentulous patients



Khalid Abushahla, Doctoral Researcher in Biomedical Engineering

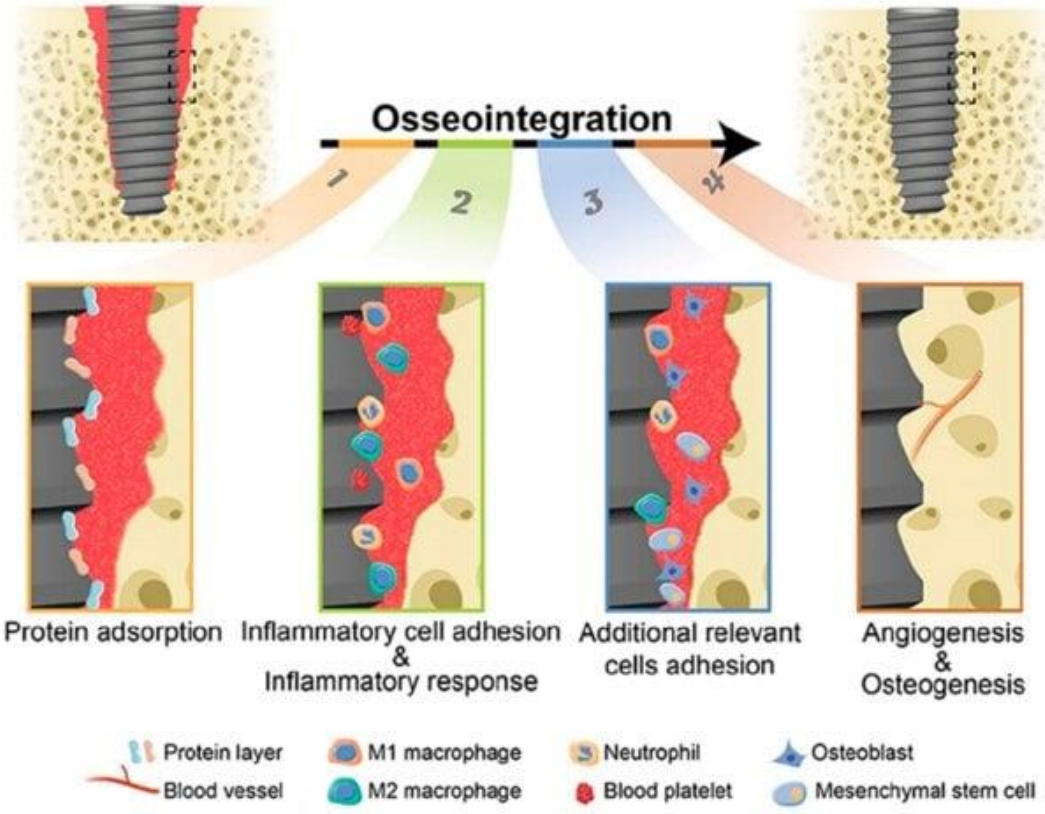
Dental Implant Materials

- Commercially available dental implants are made of:
  - Titanium
  - Titanium alloys
- Clinical success rates of placed implants have exceeded 90%.



Osseointegration

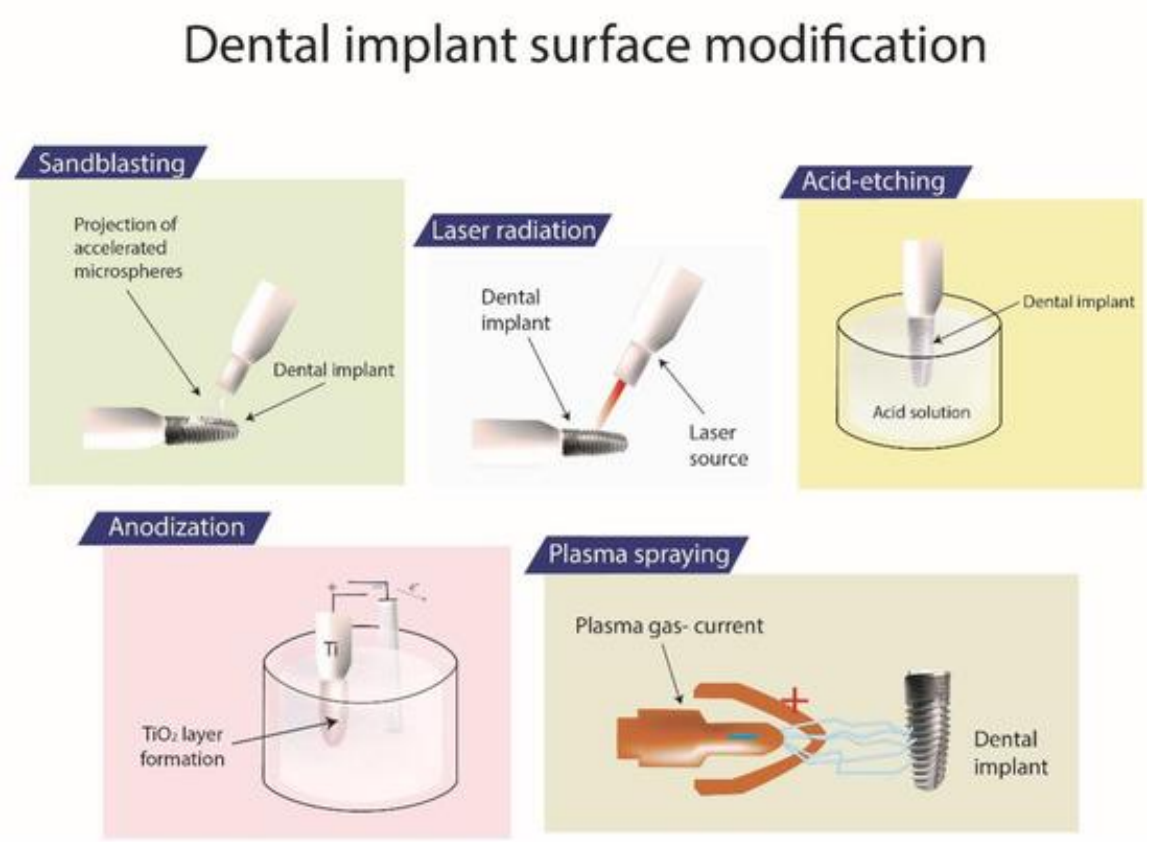
- **Osseointegration** means successful **integration between** the **implant** and **bone**.
- For **dental implants**, **successful osseointegration** allows the implant to **support crown** restorations.
- **Improved implant** designs and surface modifications help **maximize** bone-implant **osseointegration**.



Surface Modification of Dental Implants

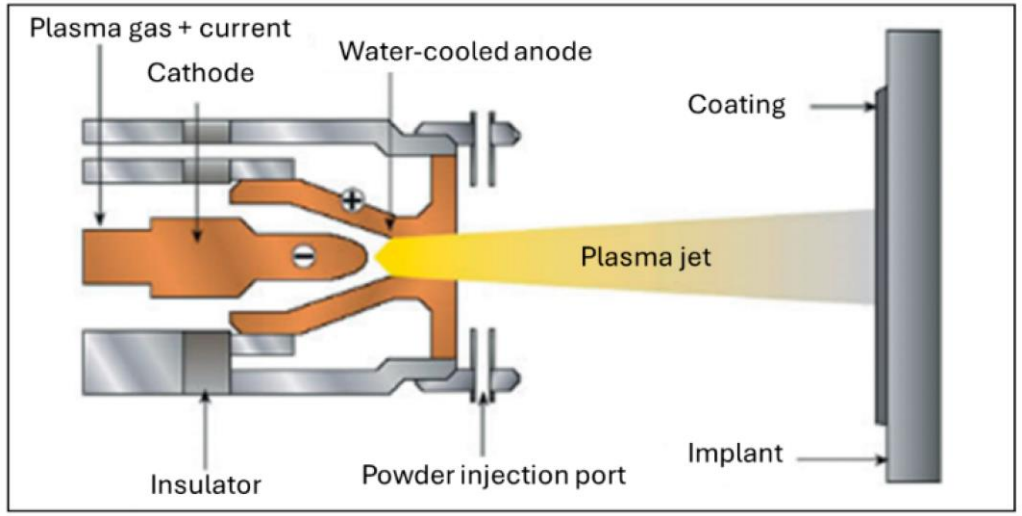
- Dental implant surfaces are **commonly modified using**:
  - Plasma spraying
  - Sandblasting
  - Acid etching

- Surface modification is used to improve bone-implant interaction.



Plasma Spraying

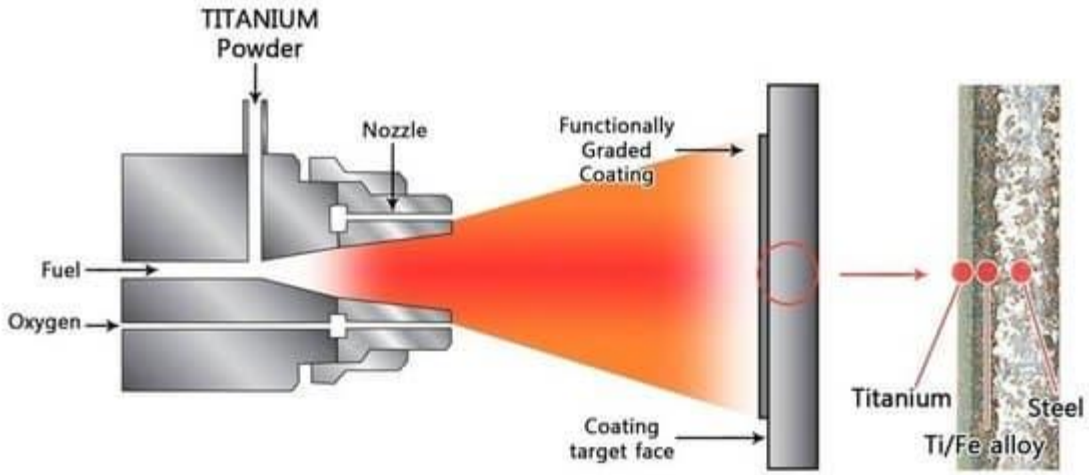
- Plasma **spraying** can **deposit** materials on **dental implant surfaces.**
- Common deposited materials include:
  - Titanium
  - Hydroxyapatite



- The implant is often sandblasted before plasma spraying to improve coating–implant adhesion strength.

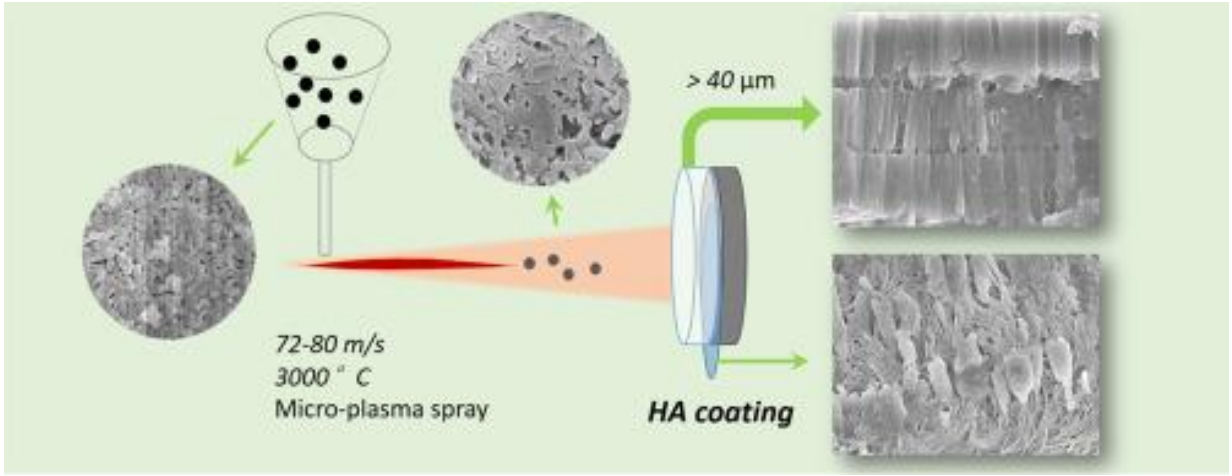
Titanium Plasma Spraying

- Titanium particles can be used in plasma spraying.
- The main rationale is to increase surface roughness.
- Higher surface roughness helps improve bone–implant osseointegration.



Hydroxyapatite Plasma Spraying

- **Hydroxyapatite** coating makes the implant surface more **bonelike**.
- **Hydroxyapatite** is **osteoconductive**.
- This can allow:
  - More rapid bone formation
  - Better implant fixation



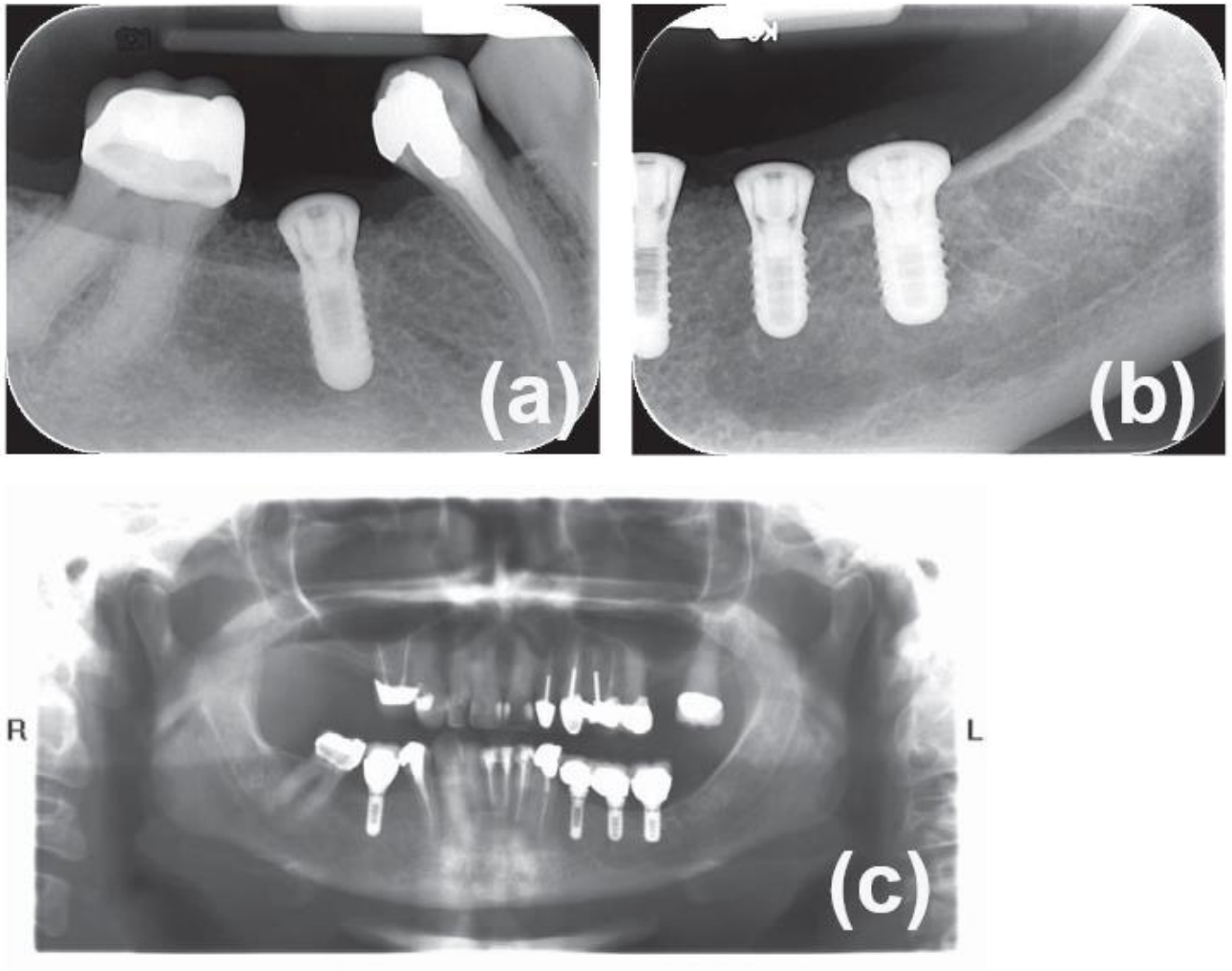


Figure 14.7

Radiographs of (a) an apical X-ray on the right mandible of a patient showing placement of one root-form implant, (b) an apical X-ray on the left mandible of the same patient showing placement of three root-form implants, and (c) a panoramic X-ray of the same patient showing implants restored with crowns after successful osseointegration. (Courtesy of Dr. Tony Dacy, Private Practice, Boerne, TX.)

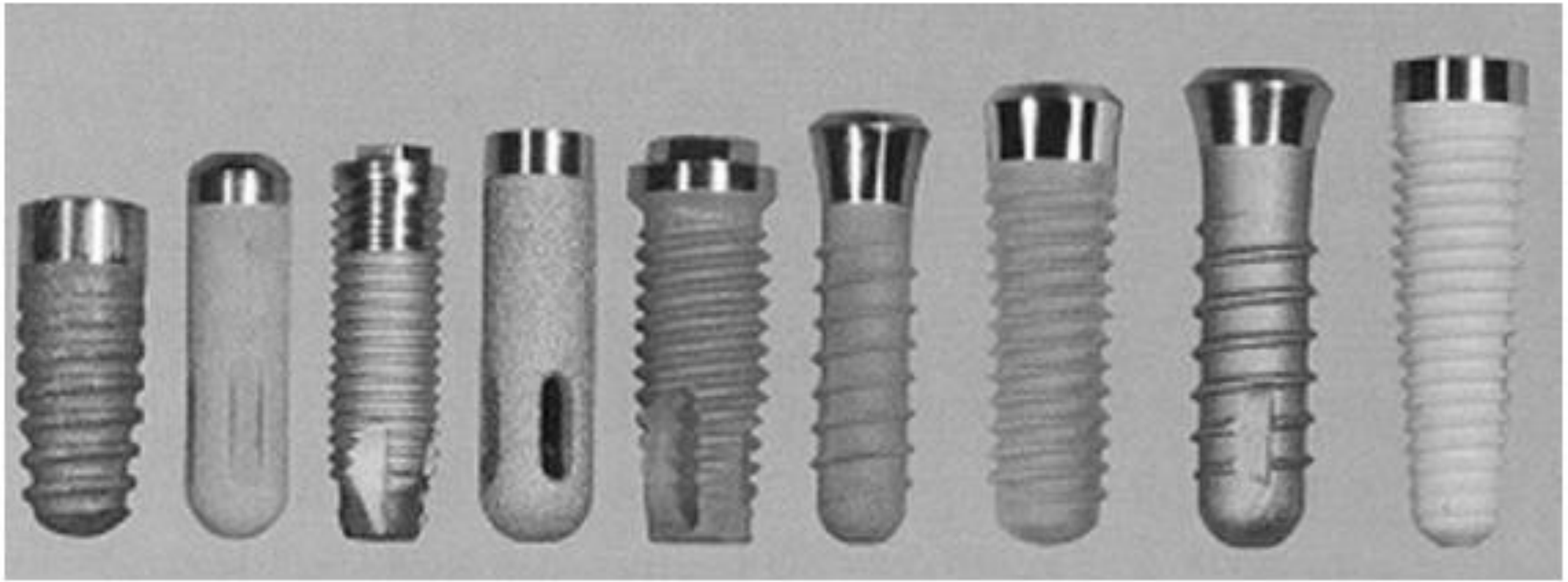


Figure 14.8

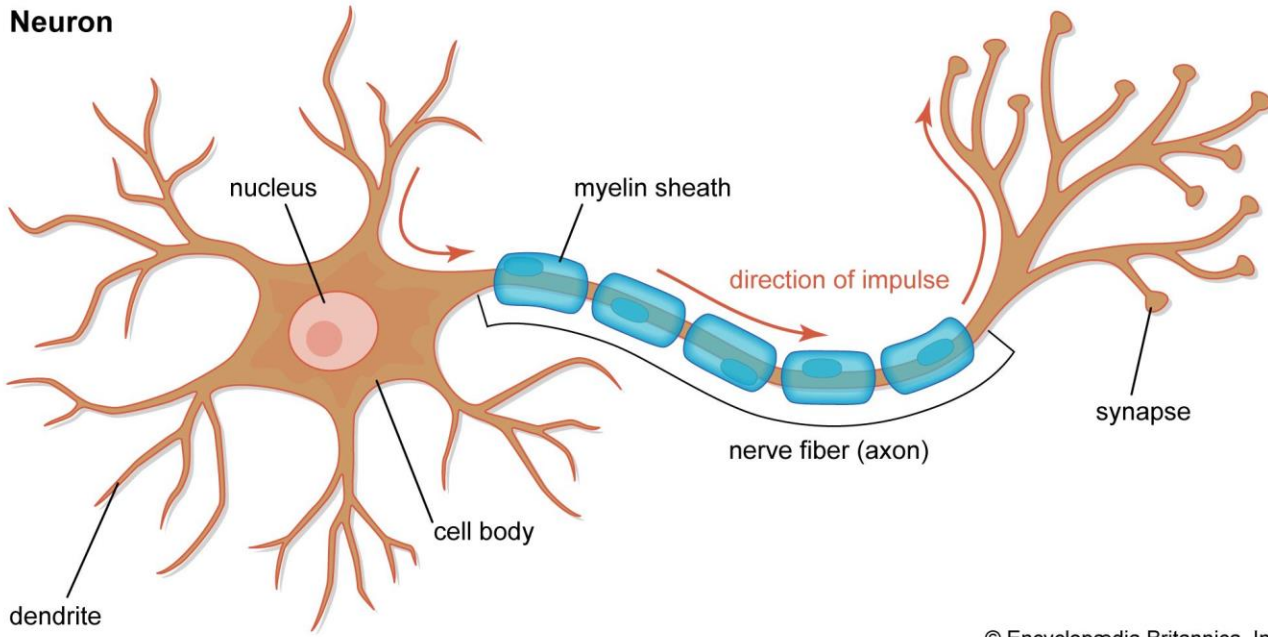
Dental implants from different vendors showing different implant design as well as different surface modifications. (Courtesy of Dr. Weihui Chen, Union Hospital, Fujian University of Medical Sciences, P. R. China.)

5. Neural prostheses

5. Neural prostheses

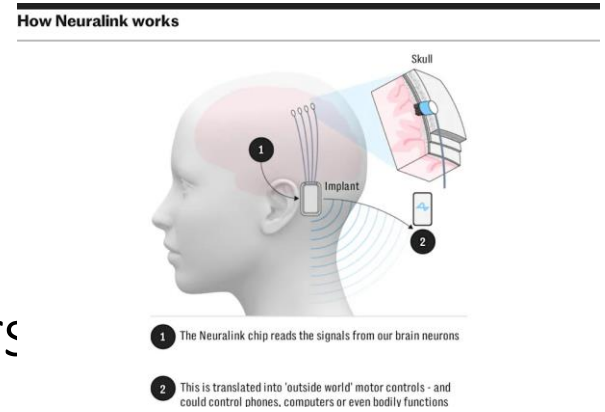
- Neural prostheses provide short **bursts** of **electrical impulses** to the **central** or **peripheral nervous** system.

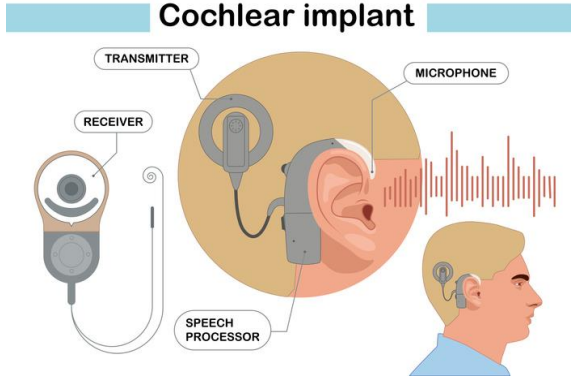
- These impulses stimulate neurons and can help produce:
  1. Sensory functions
  2. Motor functions



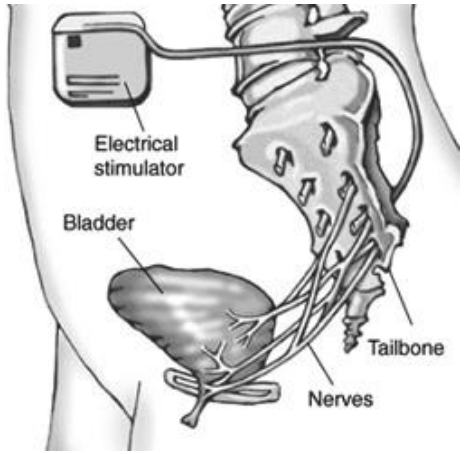
Examples of Neural Prostheses

- Common neural devices include:
  - Cochlear implants
  - Brain implants
  - Bladder management stimulators
  - Devices that help walking
  - Devices that help grasping



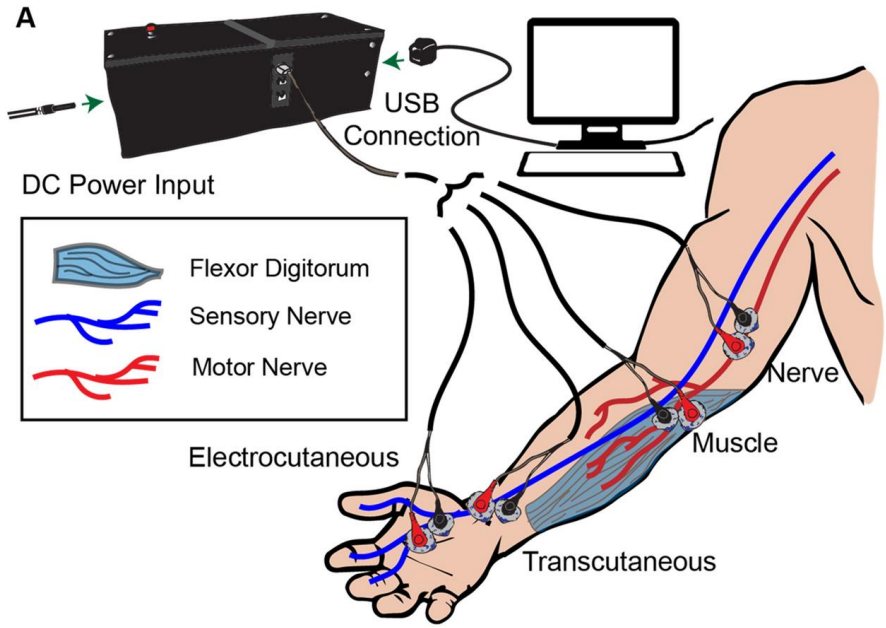






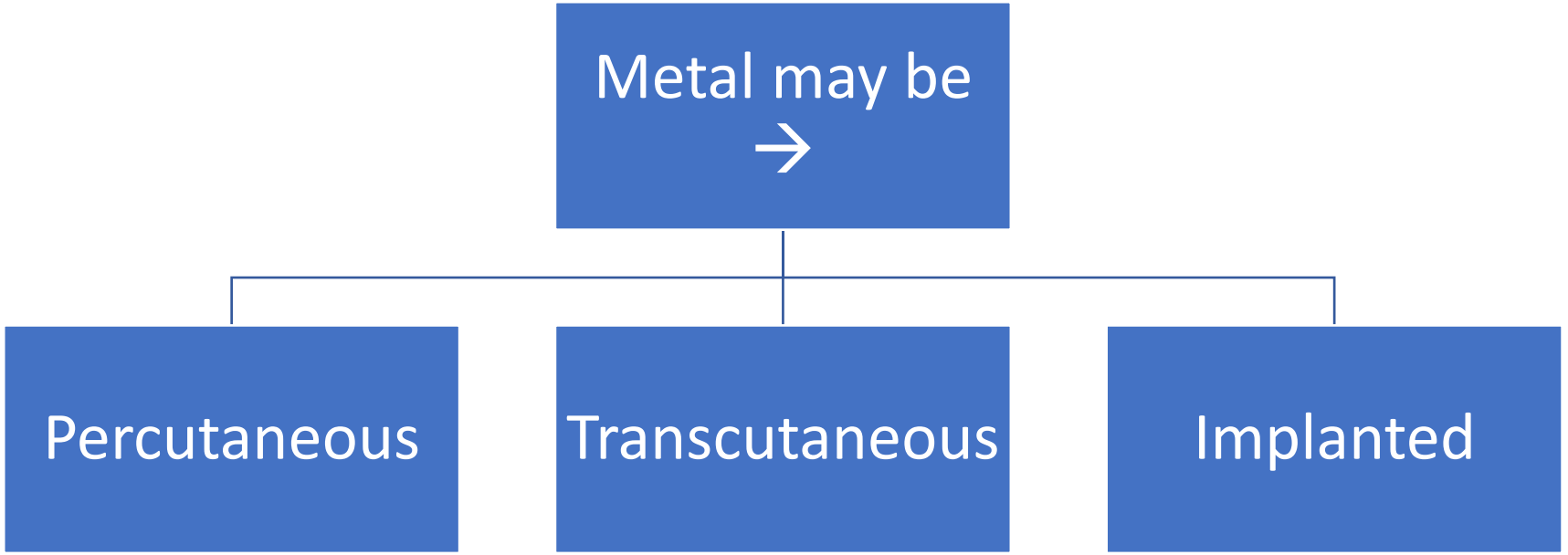
Main Components

- All neural prostheses have four common components:
  - Stimulus generator
  - Battery-powered or external AC power supply
  - Electrodes
  - User-control interface



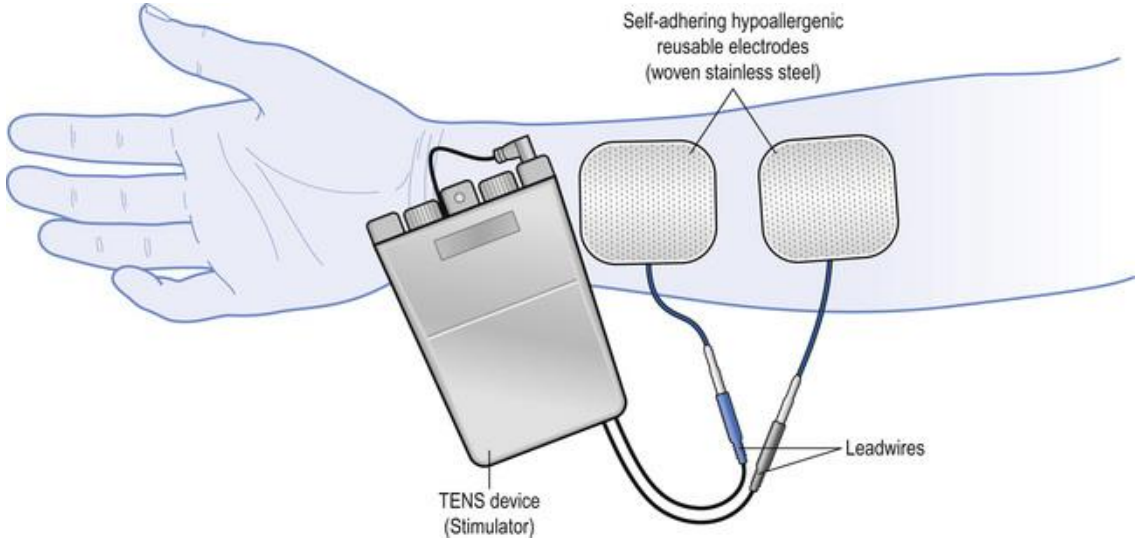
Types of Electrodes

- Electrodes' role is to stimulate nerves.



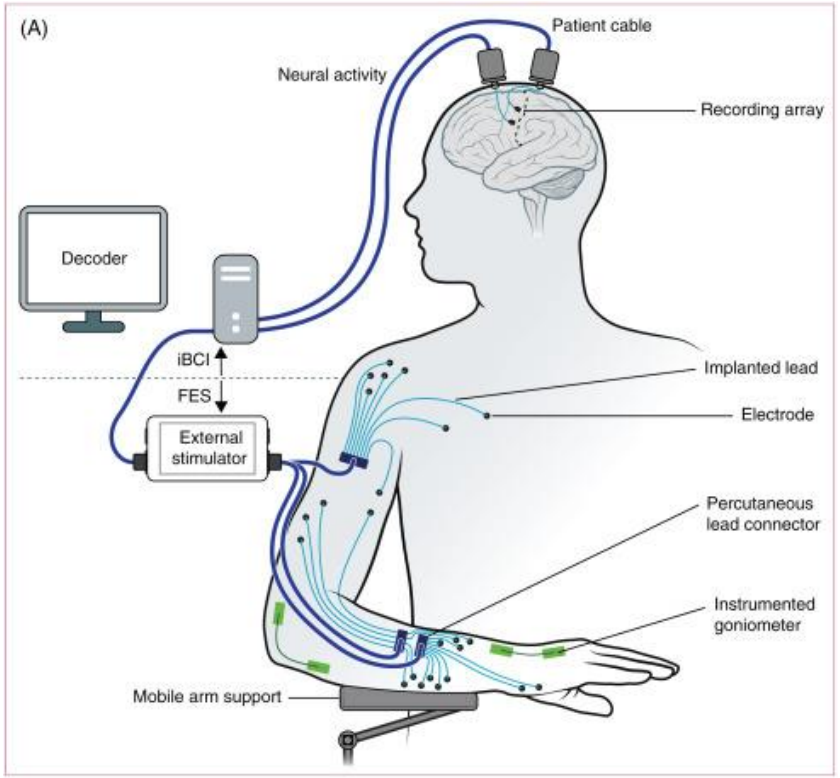
Transcutaneous Electrodes

- Transcutaneous electrodes are surface electrodes in contact with the skin.
- Advantages:
  - Noninvasive
  - Easy to apply
  - Inexpensive
- Limitation:
  - Higher signal intensity is needed because of skin impedance
    Some deep nerves cannot be stimulated effectively



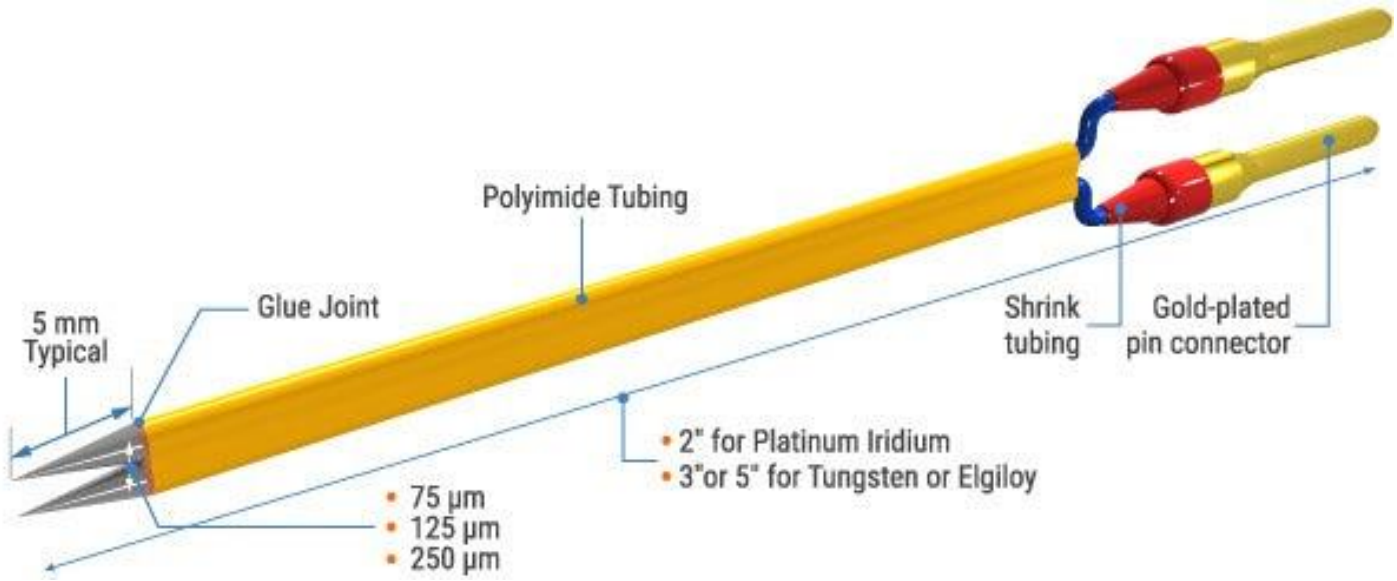
Percutaneous and Implanted Electrodes

- Percutaneous electrodes:
  - Thin wires inserted through the skin into muscular tissue
    Temporarily placed
- Implanted electrodes:
  - Permanently placed
  - Usually require a lengthy and invasive surgical process



Electrode Materials

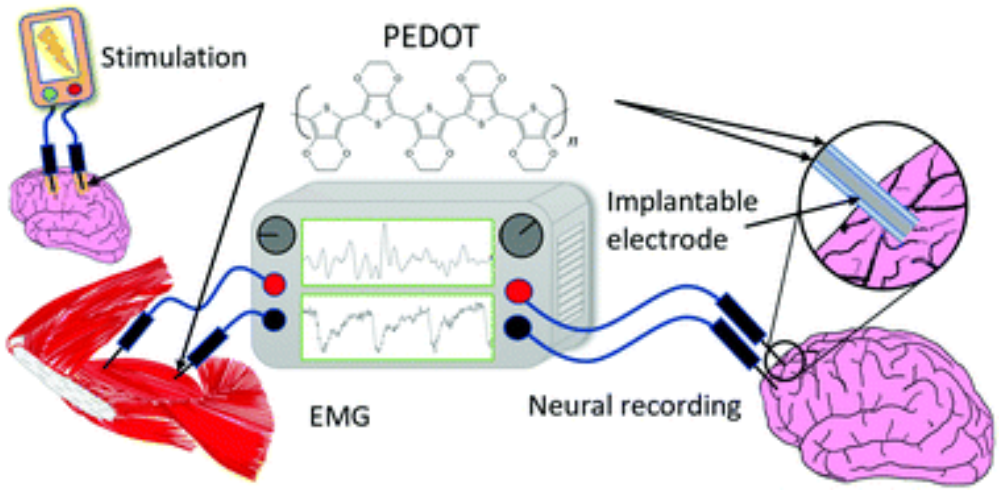
- Neural electrodes are made from metals such as:
  - Tungsten
  - Iridium
  - Platinum
  - Gold
  - Stainless steel



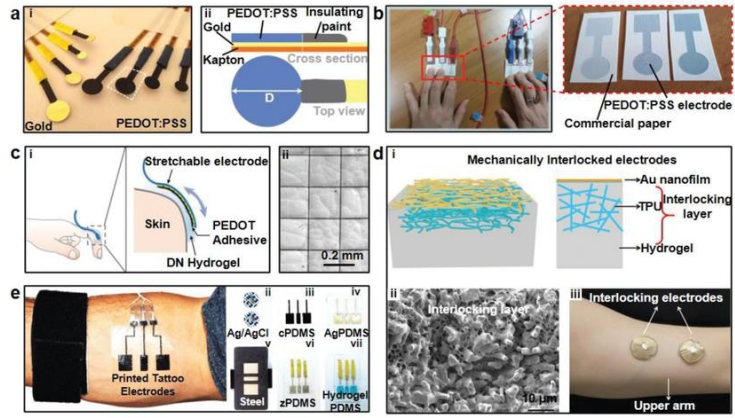
- These metals are used because electrodes must conduct electrical signals.

Advanced Electrode Coatings

- Neural signals can be improved by **coating conductive electrodes** with **biocompatible polymeric** coatings.
- Examples mentioned:
  - Alginate hydrogel
  - PEDOT nanotubes



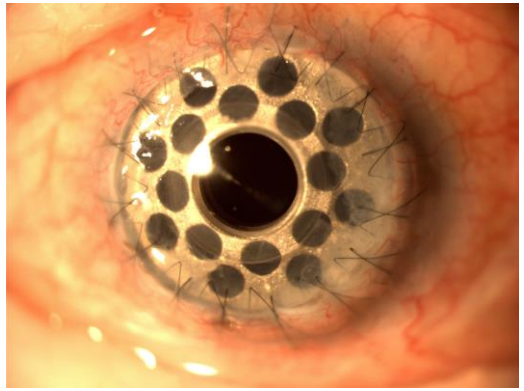
- **PEDOT means poly(3,4ethylenedioxythiophene).**



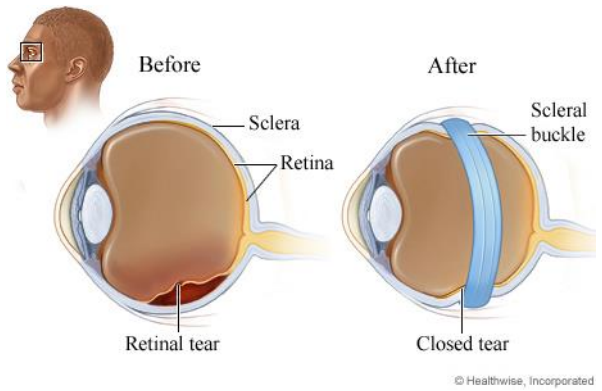
6. Ophthalmology

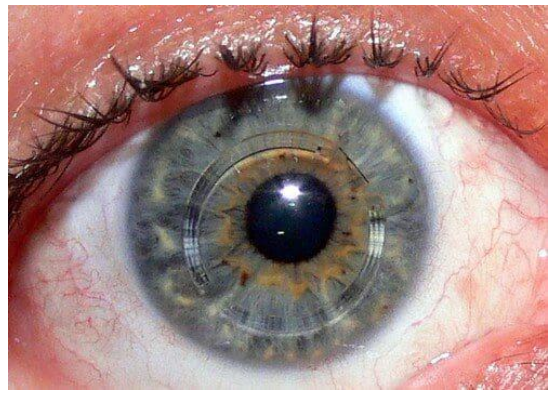
6.Ophthalmology Applications

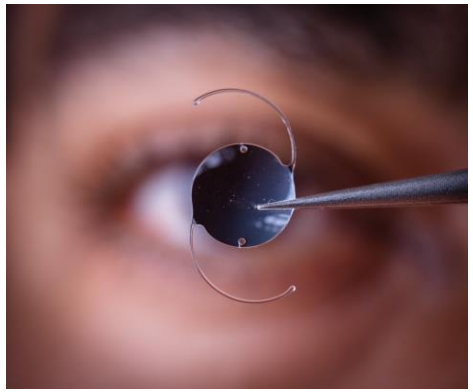
- Ophthalmic biomaterials are used mainly for:
  - Orbital implants
  - Intracorneal lenses
  - Scleral buckles
  - Glaucoma implants
  - Intraocular lenses
- Common materials:
  - Polymers
  - Ceramics

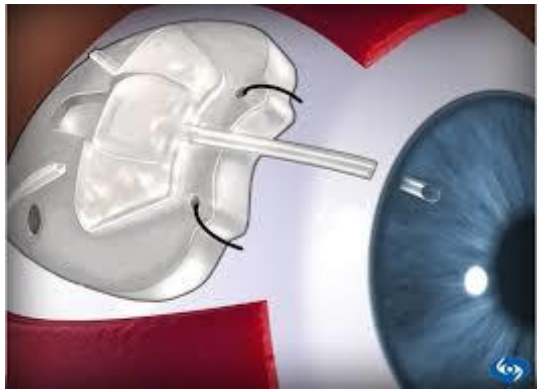












Biomaterial Selection in the Eye

- Materials are selected based on:
  - Biocompatibility
  - Optical properties
  - Mechanical stability
  - Low irritation/inflammation
- Some implants may also help deliver drugs inside the eye.

Orbital Implants

- Used after eye loss due to trauma, tumor, or severe disease.
- Main purpose:
  - Restore eye socket shape
  - Improve cosmetic appearance
- Materials:
  - PMMA or glass → non-integrated implants
  - Porous polyethylene
  - Hydroxyapatite
  - Aluminum oxide



Biointegrated Orbital Implants

- **Biointegrated** implants are **porous**.
- **Porosity allows:**
  - Tissue ingrowth
  - Vascularization
  - Better movement with surrounding tissues
- **Common materials:**
  - Coralline hydroxyapatite
  - Synthetic hydroxyapatite
  - Porous polyethylene
  - Aluminum oxide



Corneal and Intraocular Implants

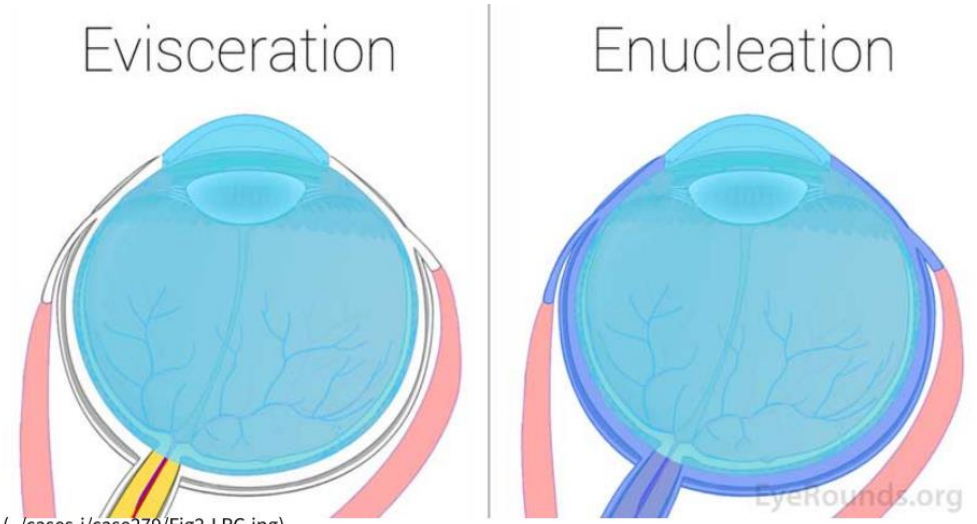
- Intracorneal lenses
  - Implanted in corneal stroma
  - Used for myopia and hyperopia
  - Materials: PMMA, hydrogel, polysulfone
- Intraocular lenses
  - Replace the natural lens after cataract surgery
  - Materials: PMMA, silicone, acrylic polymers
- Other examples
  - Scleral buckles: silicone
  - Glaucoma implants: silicone or polymer tubes/bodies

Eye Removal Procedures

- The diseased or injured eye may be removed by:

- **Enucleation:**
  - Removal of the eyeball

- **Evisceration:**
  - Removal of the eye contents
  - The sclera or shell of the eye remains



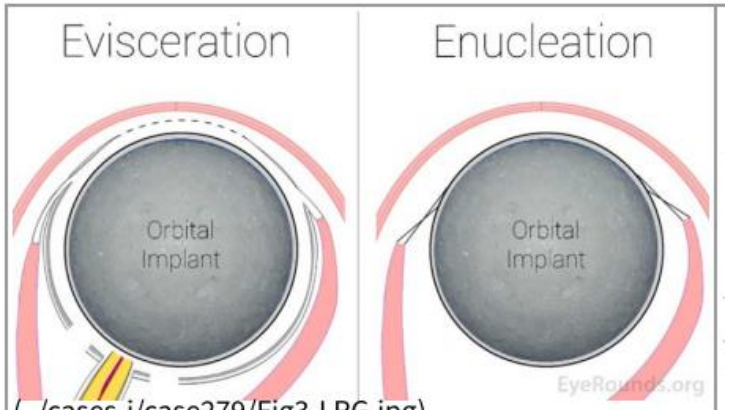
(../cases-i/case279/Fig2-LRG.jpg)

Figure 2a. *This graphic highlights the tissues removed during an evisceration.*

An evisceration removes the intraocular contents (light blue) while preserving the remaining scleral shell, extraocular muscle attachments, and surrounding orbital tissues.

Figure 2b. *This graphic highlights the tissues removed during an enucleation.*

In contrast, an enucleation detaches the extraocular muscles and removes the entire eyeball – both the intraocular contents (light blue) and the scleral shell (dark blue).



(../cases-i/case279/Fig3-LRG.jpg)

Figure 3. *This graphic depicts the placement of the orbital implant into the eye socket after the patient's has been removed via either evisceration or enucleation.*

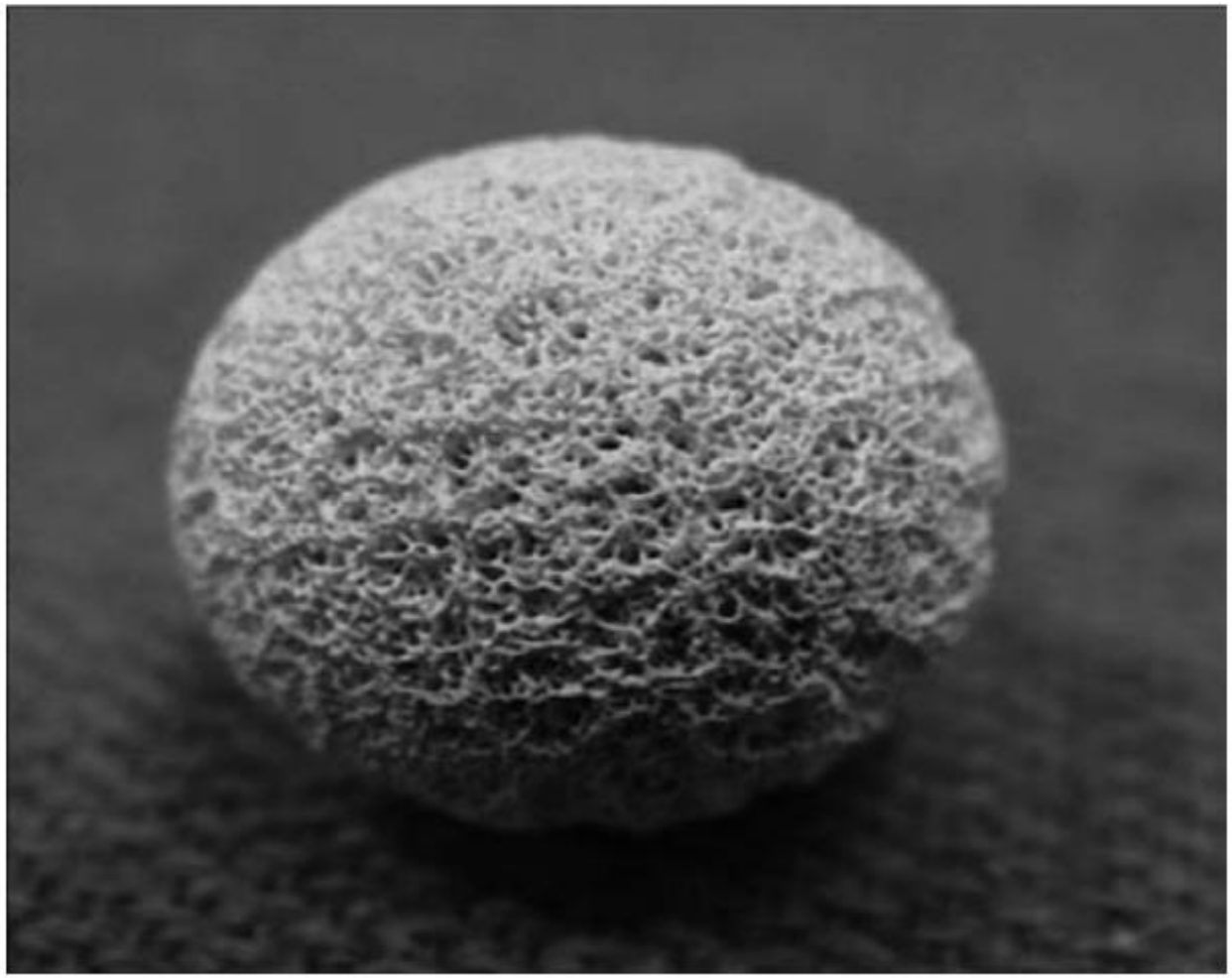


Figure 14.9

Bio-Eye® orbital implant made from hydroxyapatite. (Reproduced with permission from Integrated Orbital Implants, Inc., San Diego, CA, USA.)

Table 14.2 Commercially available orbital implants currently in the market

Biomaterial	Product	Manufacturer
Coralline HA	Bio-Eye®	Integrated Orbital Implants Inc., San Diego, CA, USA
Bovine HA	Molteno M-Sphere	IOP Inc., Costa Mesa, CA, USA
Synthetic HA	FCl_3 HA Implant	FCI Ophthalmics Inc., Issy-Les-Moulineaux, Dedex, France
Synthetic HA	Hydroxyapatite Sphere	Altomed Limited, Boldon, Tyne and Wear, UK
Aluminum oxide	Bioceramic Implant	FCI Ophthalmics Inc., Issy-Les-Moulineaux, Dedex, France
PMMA	Acrylic (Lucite) Sphere	Altomed Limited, Boldon, Tyne and Wear, UK
Silicone	Silicone Orbital Implants (Eye Sphere)	FCI Ophthalmics Inc., Issy-Les-Moulineaux, Dedex, France
Silicone	Silicone Sphere	Altomed Limited, Boldon, Tyne and Wear, UK
Porous polyethylene	Medpor™	Porex Surgical Inc., Cooledge Park, GA, USA

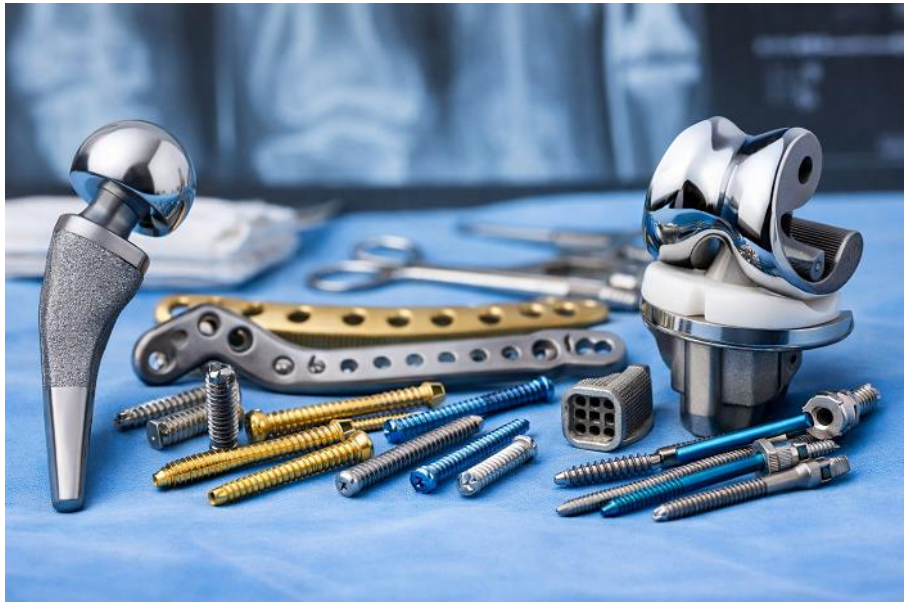
7. Orthopedic implants

7. Orthopedic implants

- Most orthopedic implants are made of:
  - Stainless steel
  - Cobalt–chrome alloy
  - Titanium or titanium alloys

- Examples include knee, total hip, finger prostheses, intramedullary nails, plates, and screws.



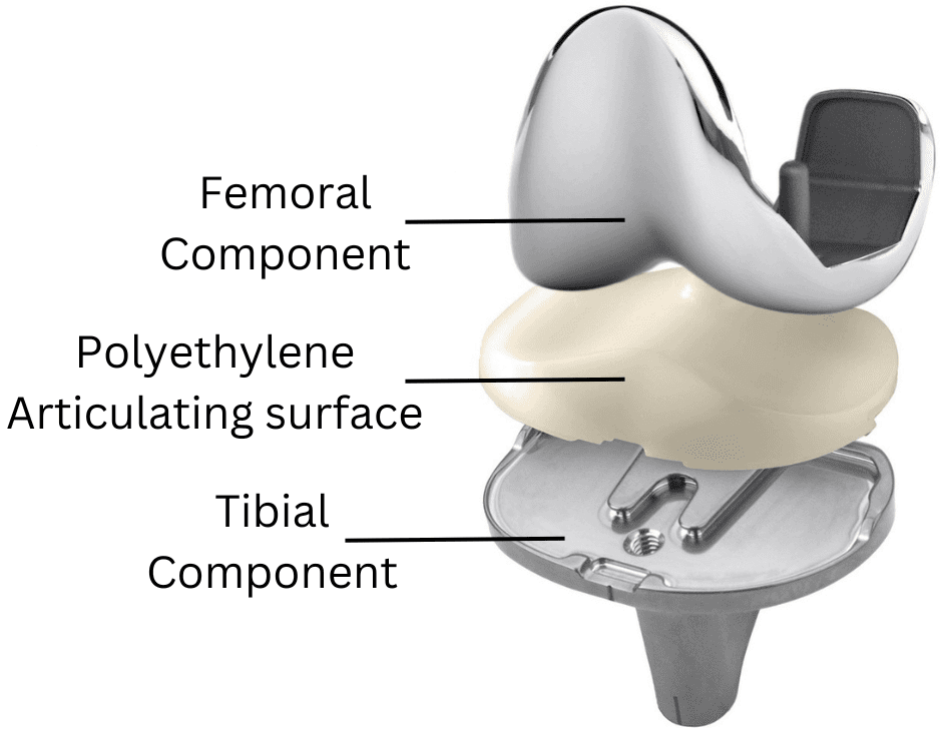


Articulating Surfaces



- When orthopedic prostheses have articulating surfaces, polyethylene polymers are used.

- In total hip prostheses, the ball is commonly made from:
  - Cobalt–chrome alloy
  - Ceramic
- These are more wear resistant than titanium or titanium alloys.



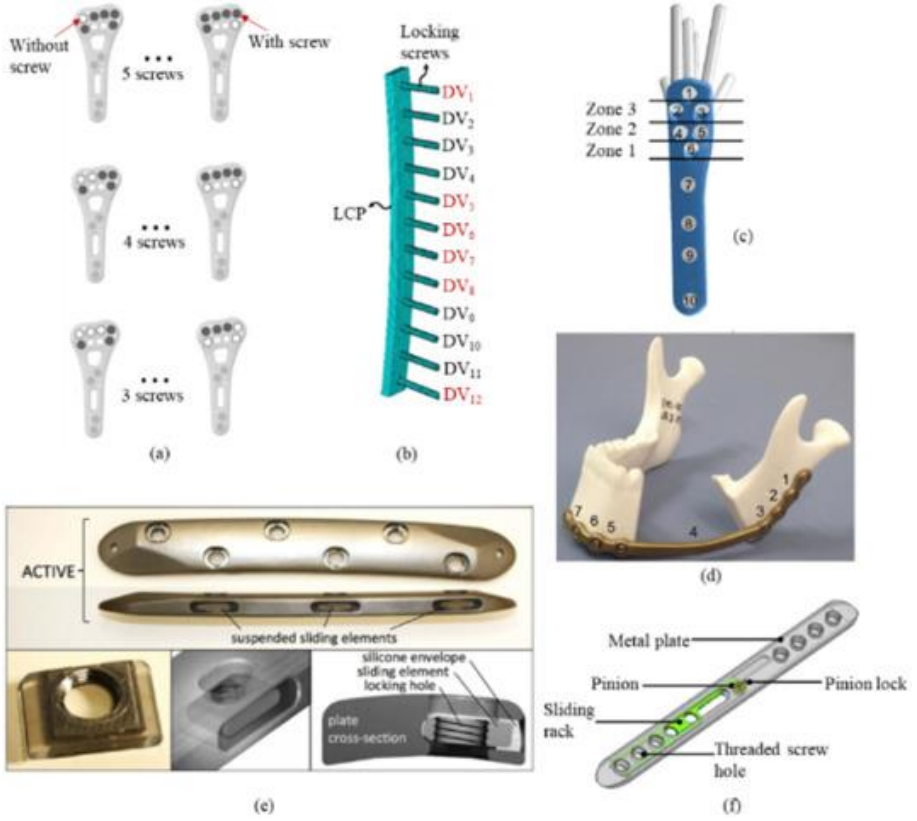
Surface Modification

- Like dental implants, orthopedic implant surfaces can be modified to enhance osseointegration.

- Common modifications include:
  - Surface roughening
  - Plasma-sprayed hydroxyapatite
  - Titanium bead coatings followed by hydroxyapatite coating

Internal Fixation Devices

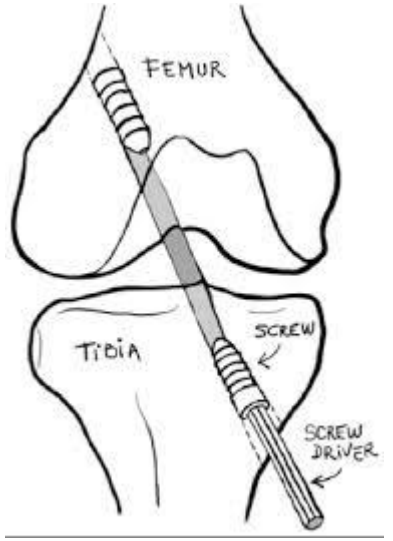
- **Internal fixation** devices are used for **fracture fixation**.
- Examples include:
  - Fracture plates
  - Metallic screws
  - Plate screws
  - Interference screws
- They are commonly made from **stainless steel** or **titanium alloys**.

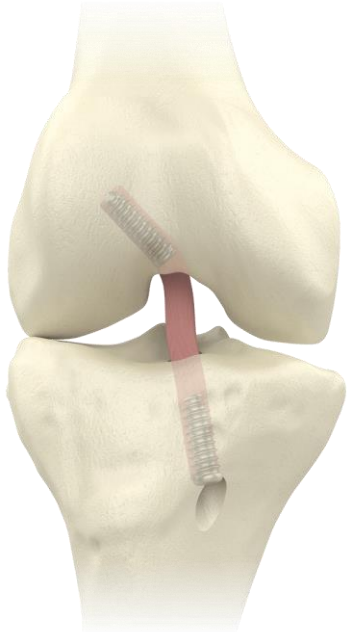


Interference Screws



- Interference screws are used to **anchor implants** or **soft tissue** to **bone**.
- Common applications include:
  - Anterior cruciate ligament, ACL, repair
  - ACL reconstruction
  - Bone–tendon graft fixation
  - Tendon reattachment to bone

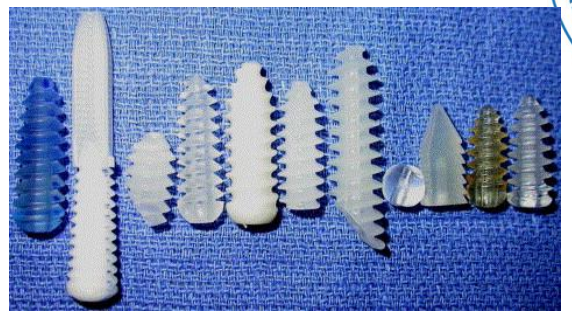




- They are placed with an interference fit at the insertion site.

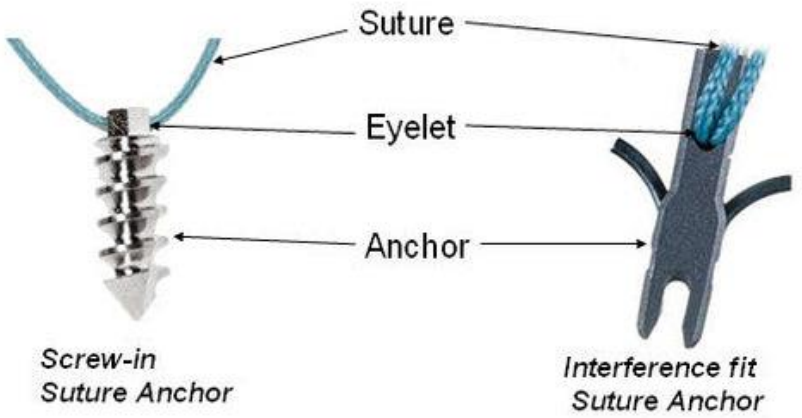
Absorbable Orthopedic Devices

- Absorbable orthopedic devices include:
  - Interference screws for ACL repair
  - Staples for meniscal repair
  - Suture anchors
  - Tacks
  - Staples for soft tissue fixation to bone









Absorbable Orthopedic Materials

- Internal fixation devices can also be made from absorbable polymers.
- Examples include:
  - PLA
  - PGA
  - PLA–PGA copolymers
  - Polydioxanone, PDO
  - Polyglyconate

- These materials can be used for pins, screws, and plates.



Figure 14.10

Synergy™ hip implant with (a) alumina femoral heads, and (b) porous bead coating followed by hydroxyapatite coating (arrow) on the proximal end of the stem. (Reproduced with permission from Smith and Nephew, Inc., Orthopaedic Reconstruction and Trauma, Memphis, TN, USA.)



Figure 14.11

Photographs of the (a) Synergy™ total hip, (b) magnified view of femoral head in the acetabular cup, and (c) Legion™ CR total knee systems. These photographs showed components made of OXINIUM™. (Reproduced with permission from Smith and Nephew, Inc., Orthopaedic Reconstruction and Trauma, Memphis, TN, USA.)

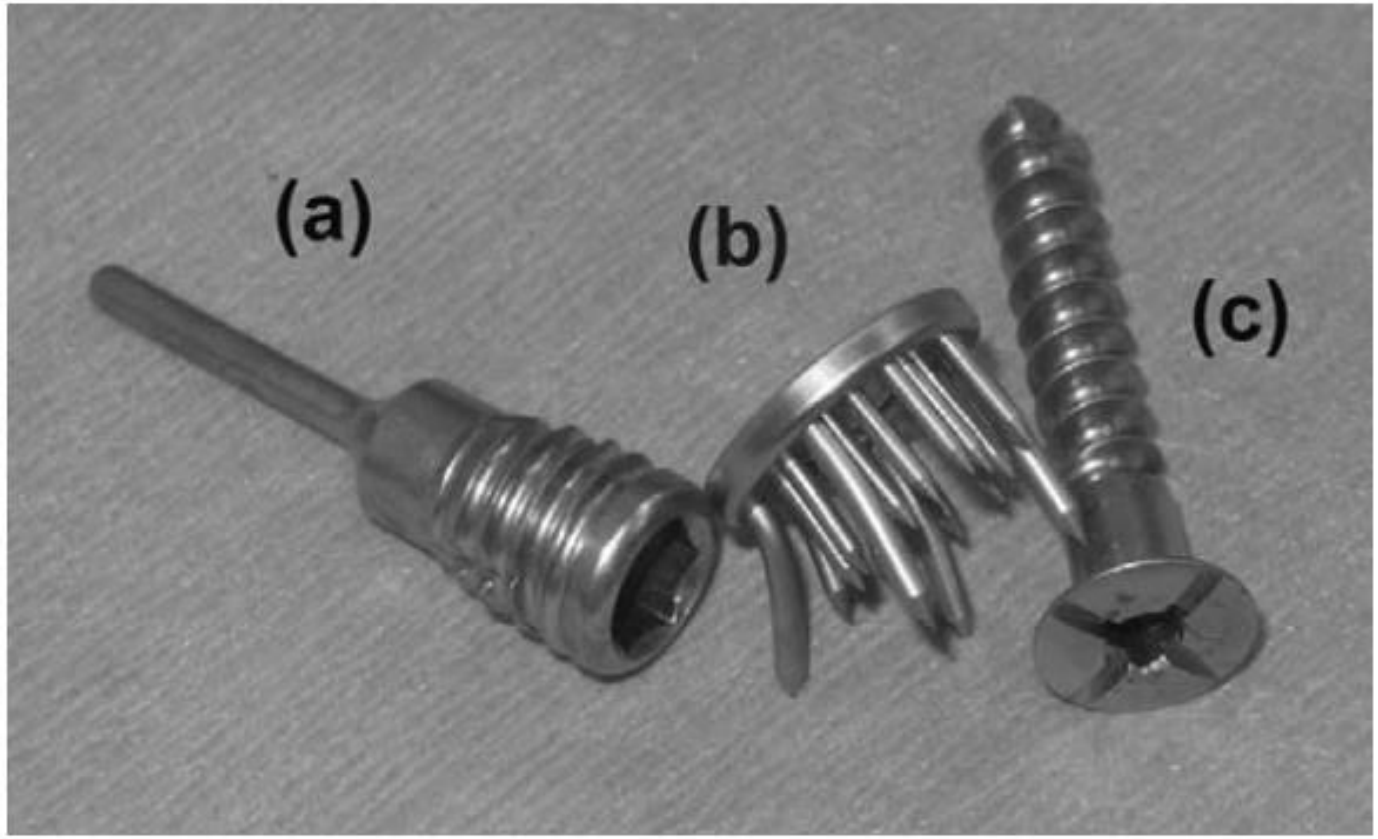


Figure 14.12

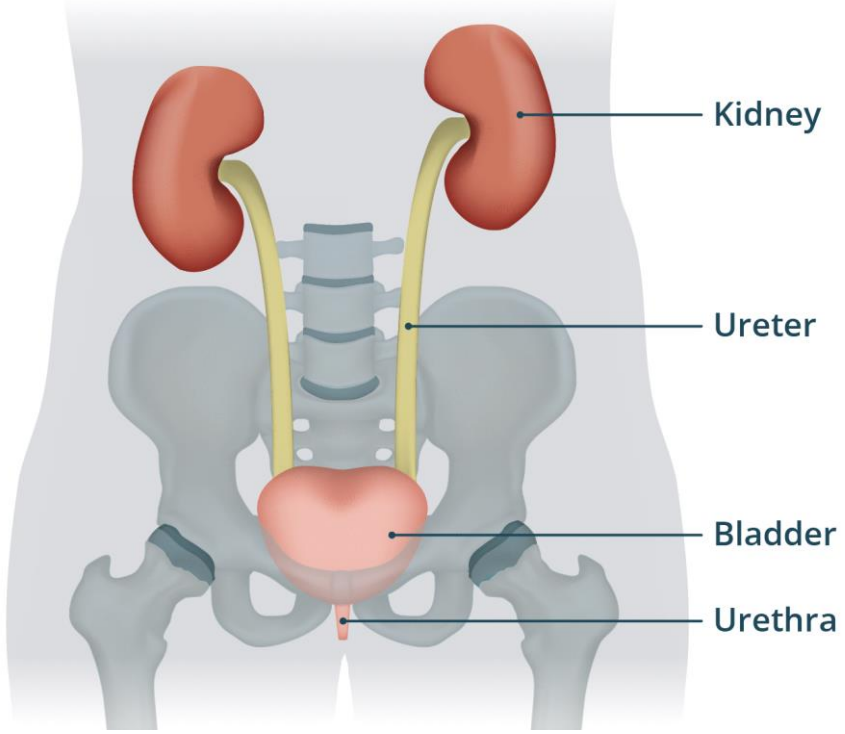
Interference screws such as the cannulated Bone Mulch™ Screw (a) for femoral tunnel fixation, a WasherLoc™ tibial fixation device (b), and a cancellous compression screw (c) that are made of titanium alloy (Arthroteck®, a Biomet, Inc. Company, Warsaw, IN, USA). These are used for fixing of graft construction during the restoration of the anterior cruciate ligament.

8. Renal

8. Renal

- **Kidneys remove**:
  - Metabolic end-products
  - Solutes
  - Excess water

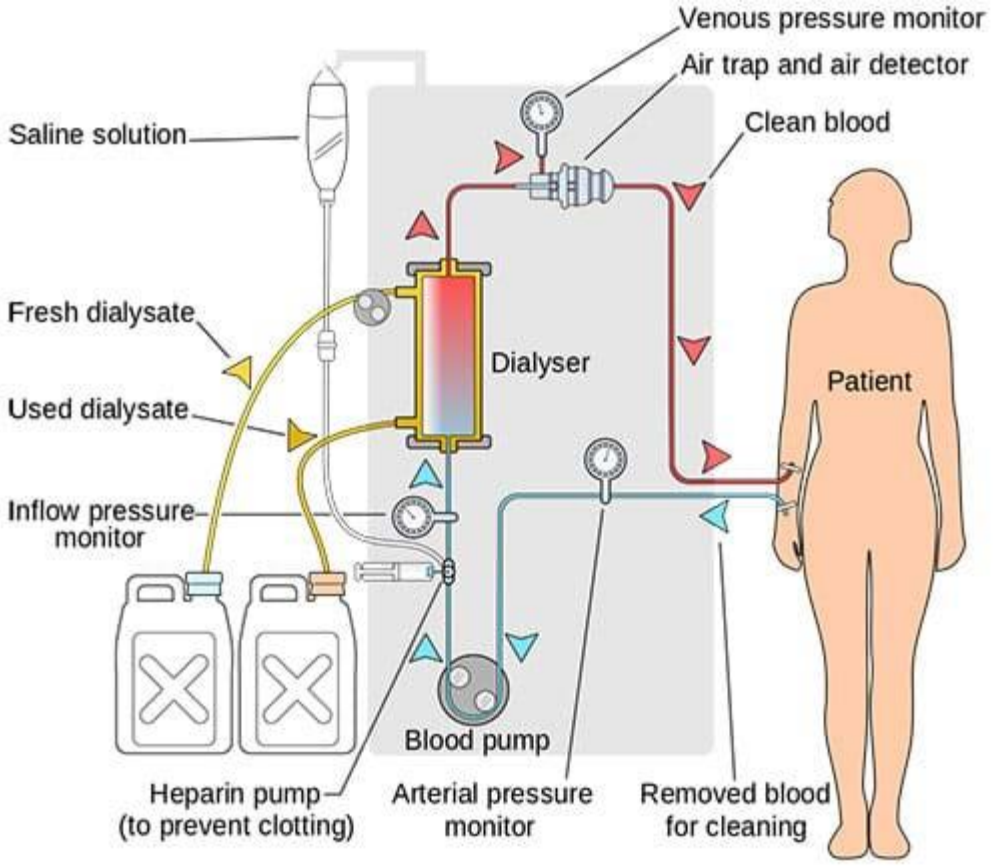
- If failed kidneys are **not** treated, solutes accumulate in the blood and may lead to death.



Renal Replacement

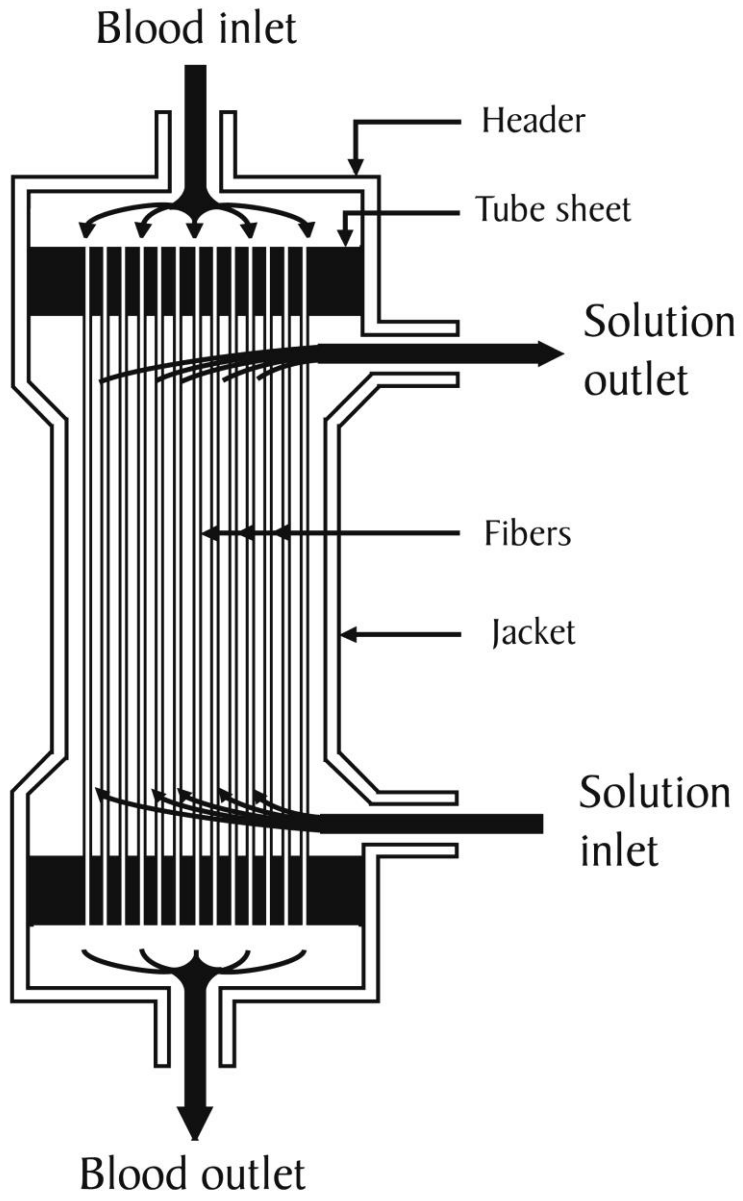
- Patients with **kidney failure** may need:
  - Renal replacement therapy
  - Kidney transplantation

- **Dialysis therapy** depends strongly on **semipermeable membranes**.



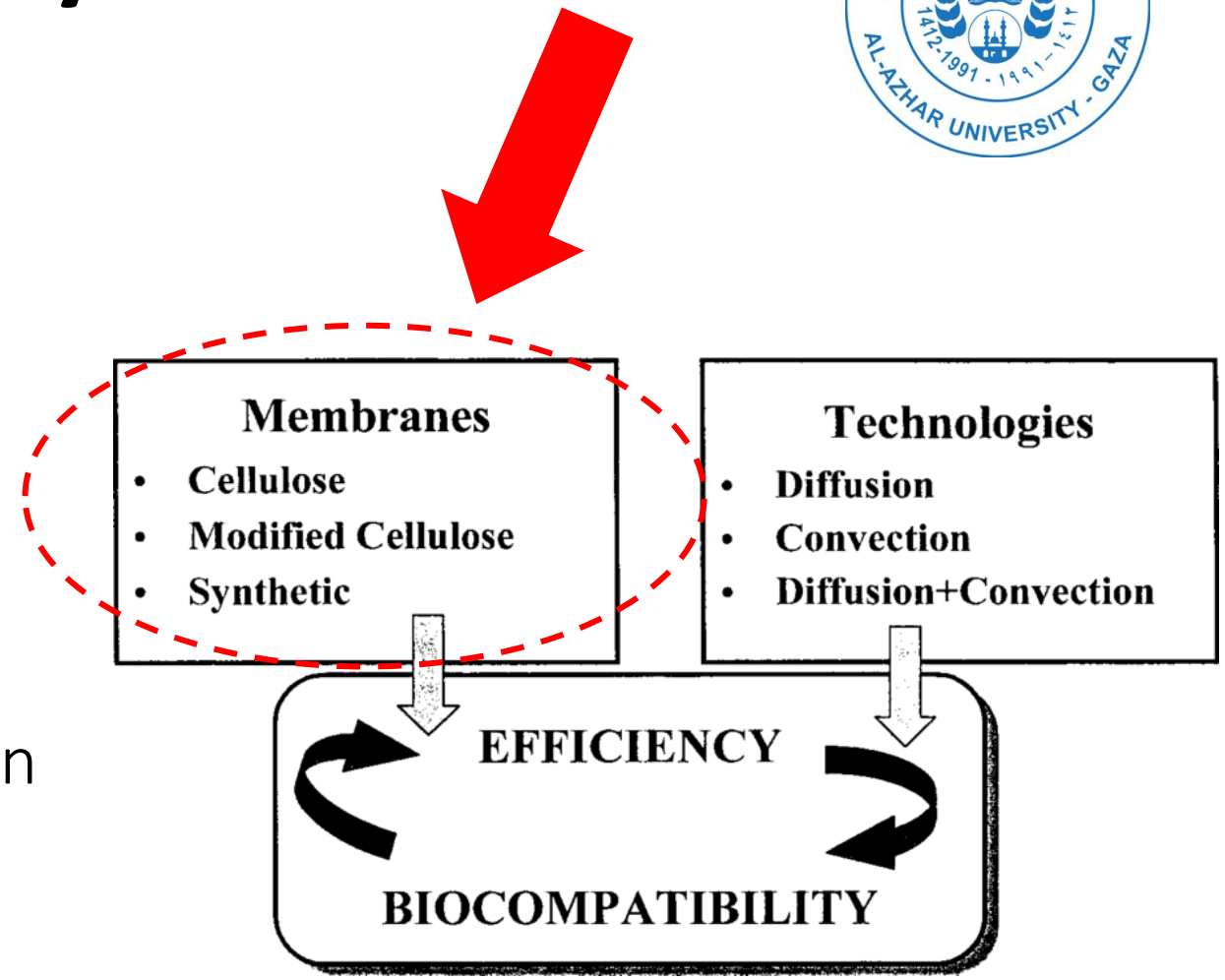
Ideal Dialysis Membranes

- Ideal dialysis membranes should:
  - Remove a broad range of metabolic end-products
  - Be biocompatible
  - Resist changes during sterilization
  - Not cause adverse reactions with blood
  - Not react negatively with administered drugs



Biocompatibility in Dialysis

- In dialysis membranes, biocompatibility means:
  - No adhesion or activation of blood cells
  - No adsorption of proteins
  - No transformation of proteins
  - No transformation of nonprotein substances by the membrane



I-First Generation Dialysis Membranes

- First generation dialysis membranes are made from **regenerated cellulosic** membranes.
- **Cellulose** is a **natural linear polysaccharide**.
- These membranes are:
  - Highly hydrophilic
  - Low in water and solute permeability
  - Small in pore size
  - Thin, about 5–15 µm
- Examole : Cuprophan®

II-Modified Cellulosic Membranes

- Modified cellulose membranes were developed to improve biocompatibility.
- Through acetylation, these membranes do not bind strongly to C3 molecules.
- This decreases complement activation.
- Example:
  - Hemophan®

III-Synthetic Polymer Dialysis Membranes

- Dialysis membranes can also be made from synthetic polymers such as:
  - Polyacrylonitrile
  - AN69
  - Polysulfone
  - PMMA
  - Polyamide
- AN69 is a copolymer of acrylonitrile and sodium methallyl sulfonate.

Synthetic Membrane Properties

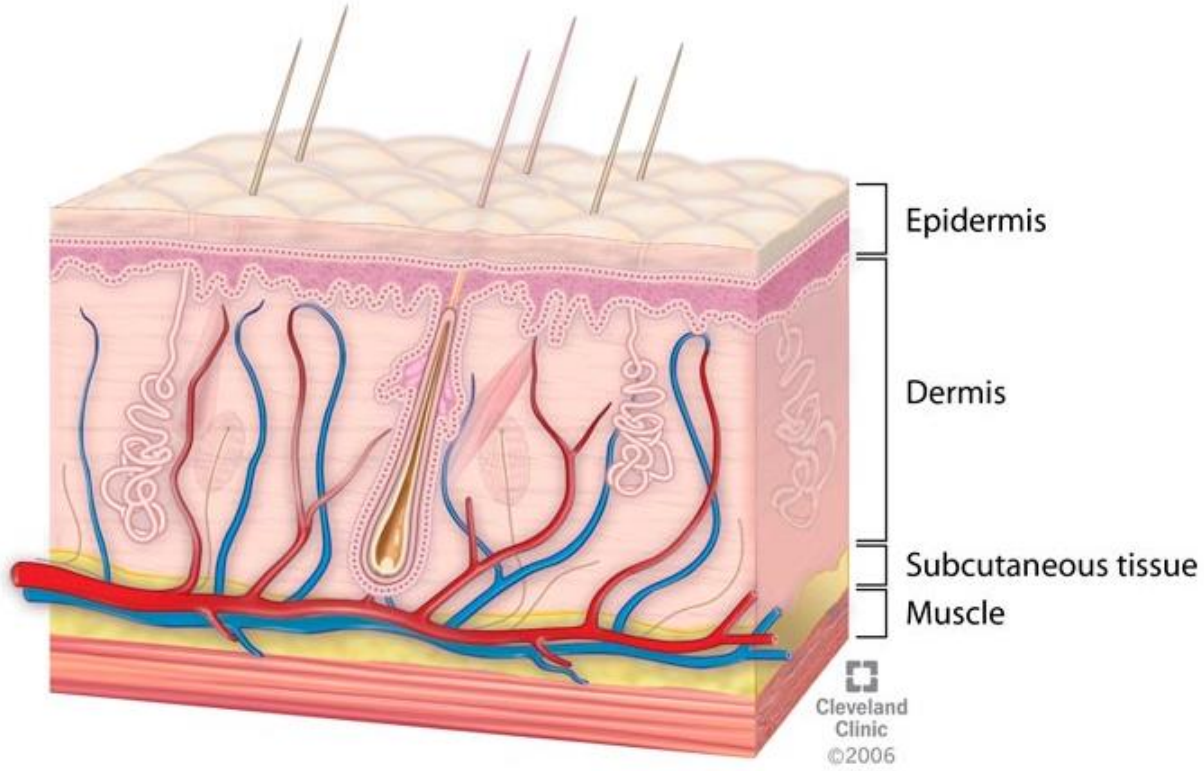
- Synthetic polymer membranes are generally:
  - More porous
  - More hydrophobic
  - Higher in protein adsorption capacity
  - Thicker, at least 20 µm
  - More biocompatible
- Their increased biocompatibility explains their growing popularity.

9. Skin Applications

Skin Applications

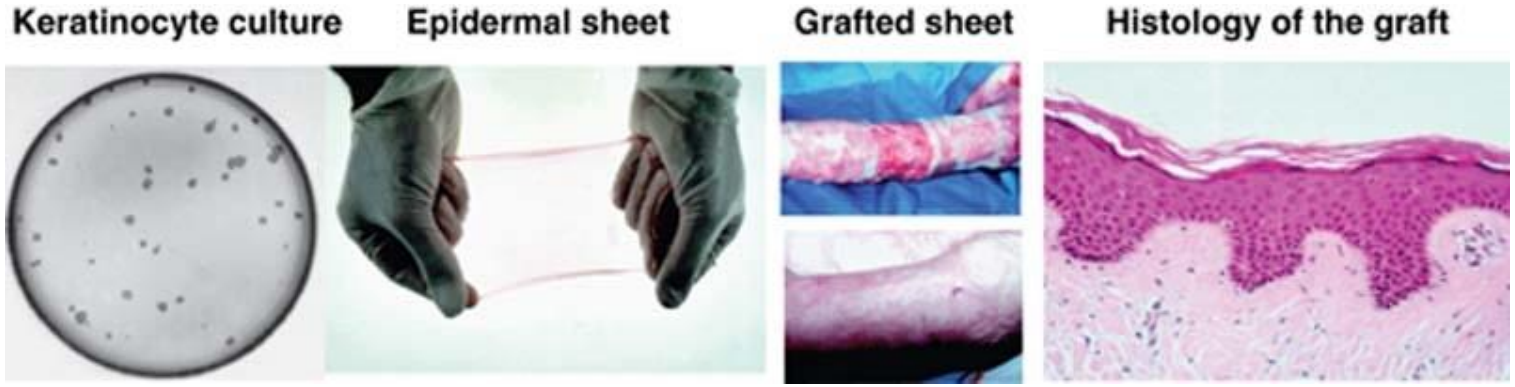
- Skin protects the body from:
  - Infection
  - Water loss
  - Electrolyte loss

- Severe burns or wounds may require skin substitutes or dressings.



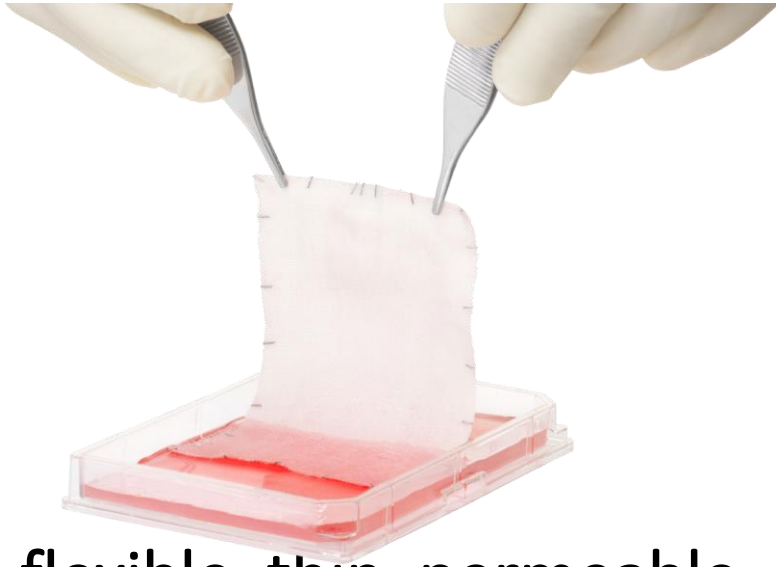
Cultured Epidermal Autografts

- Used for severe burn patients.
- Prepared from the patient's own skin cells.
- Main use:
  - Replace damaged epidermis
- Cover large burn areas
- Examples:
  - Epicel®
  - Holoderm®



Biomaterial Films for Cell Transfer

- **Thin films** help transfer **cultured epithelial** cell sheets to **wounds**.
- Important materials:
  - Collagen
  - Hyaluronan
  - Chitin
  - Fibronectin
  - Fibrin



- Ideal films should be **biocompatible**, **flexible**, **thin**, **permeable**, and **biodegradable**.

Temporary Skin Substitute Dressings

- Used when direct grafting is difficult or delayed.
- Main functions:
  - Protect the wound
  - Reduce water loss
  - Prevent infection
  - Control bleeding
- Materials:
  - Silicone
  - PVC membranes
  - Polyurethane membranes
  - Silicone-coated gauze or nylon mesh





Interactive Skin Substitutes



- These materials interact with tissue and help healing.
- Examples:
  - Dermagraft®: fibroblasts on polyglactin mesh
  - Biobrane®: silicone film + nylon fabric
  - Integra®: silicone layer + bovine collagen/GAG matrix
- Main use:
  - Support dermal regeneration
  - Temporary or long-term wound coverage



Figure 1. Apligraf (A) and Dermagraft (B) as received on removal from manufacturer's packaging.

Summary

- Clinical biomaterials are designed for specific medical applications.
- Successful implants and devices depend on:
  - Material properties
  - Biofunctionality
  - Surface properties
  - Tissue response after implantation
  - Manufacturing technology
  - Biocompatibility
  - Cost and material limitations
- Often, different materials and drugs are combined to improve device performance.
- Example: Total hip prostheses may combine titanium, cobalt–chrome alloys, and UHMWPE.

Think with me 🤔🤔🤔

- Give one example of a **direct biomedical application of biomaterials** that was **not mentioned in the lecture**. Briefly explain:
  1. The application
  2. The goal of using the biomaterial
  3. The biomaterial used
  4. Why this biomaterial is suitable for this application

Thank you